The two-body kernels of Q_1, S_1 and K_1: defect graphs, closed forms, renormalisation freedom and the twist check

Step 2, composite and odd two-body kernels, part Q (theory)

2026-10-30

Abstract

The one-loop components of the superconformal action on local operators along the supercharges $\xi \in \Sigma = S_+ \otimes W \oplus S_- \otimes W^*$, the special supercharges $\eta \in \Sigma^*$ and the special conformal fields K_b are read off from the flow of the defect vertex of Lemma c8-transfer. We classify its graphs along each of these generators. Every graph that meets the operator in two letters is a cycle through the pair with exactly one special edge $K_\varepsilon - K_0$ and exactly one vertex $S^{(1)}(x)$, placed on the cycle in one of three ways: with both legs on edges of $P(\varepsilon, T)$ (the composite line along x: a λ–c line along Σ, the $A|_V$–c line along K_b), with one leg on the special edge, or, for the cubic term of $S^{(1)}(\xi)$, with two legs on the cycle and an antifield output. The remaining graphs cancel against the terms of the observable at scale T, are removed by the counterterm pairs, or vanish by the one-loop master equation through arity one. Hence, for every local functional O, gauge-variant ones included, the component along x is a sum over the pairs of letters of a local two-body kernel, the ε^0 coefficient of these cycles corrected by finite special vertices, plus the one-body term $-\{\hbar F_1^{(1)}(x), -\}$. We derive the commutators of the operators $x.$ with Q, Q^{GF} and H, which produce the composite lines and the variation kernels of the propagator, prove a Ward identity for the flow along ξ, give the census of the cycles (45 of the 55 pairs of slots carry them along every generator), the output bounds with the weight count of the position-dependent Killing data, the exact closed forms as simplex integrals with Bessel generating functions, the position decomposition that reduces the parts of S_1 and K_1 multiplying the position to the kernels along Σ, the translations, the dilatation and the rotations, the independence of the endpoint for K_1 with the $\log T$ term of Definition c8-K, the complete renormalisation freedom and the invariant content, and the comparison with the twisted Q_1 of [BK, (3.7), (3.64)–(3.69)] along a rank-one supercharge. Two engine routes and their gates are specified in `engine_spec.json`.

Contents

1 Data and names **2**

2 The generators on the regulator **4**

3 The defect graphs and the two-body form of the components **8**

4 Census of the defect cycles and the output bound **13**

5 Closed forms of the two-body kernels **17**

6 Renormalisation freedom and invariant content **20**

7 The twist check and the inner-product test **22**

8 One-loop relations among the components **25**

9 Engine specification **27**

10 Two-component forms **28**

11 Status **29**

1 Data and names

All conventions are those of the manuscript, Chapters 1–11 (Sections 11.1–11.5 final), of the two-letter theory (`step2/twoletter/theory/twoletter_theory.tex`) and of the composite theory (`step2/composite_qsk/composite/theory/composite_theory.tex`). References of the form Def c8-K, Prop c11-oddbound are to labels of the manuscript. Weight $e^{+S/\hbar}$, $S=\frac12\langle e,Qe\rangle+I$; on arity zero $s=Q+\{I,-\}$; edges $P=P(\varepsilon,T)=\int_\varepsilon^T(Q^{GF}\otimes 1)K_t\,dt$ (Def c8-PT) and $P\star=\int_\varepsilon^T Q^{GF}e^{-tH}dt$ the operator it represents; the special edge is $K_\varepsilon-K_0=-\int_0^\varepsilon(H\otimes 1)K_t\,dt$, the factor $e^{-\varepsilon|p|^2}-1$ times the components of K_0 (Prop c4-defect1). No dimensional regularisation enters in any role; the only infrared datum is the transfer endpoint T, which is set to ∞ wherever the graph sums converge at generic momenta (Prop c8-Tsafe) and kept finite along K_b (Def c8-K). The unit is

$$u=\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}=-\frac{g^2N}{32\pi^2}\qquad(\mathfrak g=\mathfrak{su}(N)).$$

Generators and components. As in Chapter 8, $\xi\in\Sigma$ with two-component data $(\xi_\alpha{}^i,\bar\xi_{\dot\alpha i})$ (generators Q^i_α, $\bar Q_{\dot\alpha i}$), $\eta\in\Sigma^*$ with data $(\eta^\alpha{}_i,\bar\eta^{\dot\alpha i})$ (generators $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$), K_b the special conformal field $v(x)=(x,x)b-2(b,x)x$, P_b the translation, Δ the dilatation ($v=2E$), M_ω the rotation of V with the antisymmetric endomorphism ω. The letters $Q_1,S_1,D_1,K_1,P_1,M_1,R_1$ are the one-loop components $\hbar L_1(O)^{(1)}(x)$ of Def c8-components; Q alone is the linear BV differential. The ad_Δ eigenvalue λ_x is -1 on Σ, $+1$ on Σ^*, $+2$ for K_b, -2 for P_b, 0 for Δ and the rotations; the degree of the Killing data of x in the position is 0, 1, 2, 0, 1, 1 (Section 10.9).

Letters and pairs. Letters, Weyl weights, degrees, the pair weight $w_{X_a}+w_{X_b}$, the pair degree and the Koszul signs are those of Def c10-letters and (3.1) below, which is eq c10-decomp. The slots are c, $A|_V$, $A|_\perp:=A|_{W^\perp_{\mathbb R}}$, $\lambda_+:=\lambda|_{S_+\otimes W}$, $\lambda_-:=\lambda|_{S_-\otimes W^*}$, $A^*|_V$, $A^*|_\perp$, $\lambda^*_+:=\lambda^*|_{S_+\otimes W^*}$, $\lambda^*_-:=\lambda^*|_{S_-\otimes W}$, c^*, with Weyl weights $0,1,1,\frac32,\frac32,3,3,\frac52,\frac52,4$ and degrees $1,0,0,0,0,-1,-1,-1,-1,-2$; two-component c, A_m, A_{ij}, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$, A^*_m, A^*_{ij}, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$, c^* (Def c8-twocomp). The arity-zero counterterm of a local functional M is written

$$\partial_\delta I^{CT}[\varepsilon](M)=\sum_{m\ge1}\varepsilon^{-m}\,\mathrm{ct}_m(M)+\log\varepsilon\,\mathrm{lc}(M),$$

with ct_m, lc two-body by Thm c10-decomp; the arity-one counterterm of M along x (the singular part of the cycles through two letters of M with one vertex $S^{(1)}(x)$ on the cycle, Def c11-graphs for $x\in\{\Delta,P_b\}$, Definition 3.2 below in general) is written $\mathrm{ct}^{(1)}_x(M)=\sum_m\varepsilon^{-m}\mathrm{ct}^{(1)}_{x,m}(M)+\log\varepsilon\,\mathrm{lc}^{(1)}_x(M)$.

Definition 1.1 (The terms of $S^{(1)}(x)$)**.** For $\xi\in S_+\otimes W$ the functional $S^{(1)}(\xi)=\int\mathrm{dvol}\big(-(\Gamma(\xi,\lambda),A^*)+\frac12(\rho(F_A)\xi,\lambda^*)\big)$ (eq c2-S1Q) is the sum of the seven terms of Table 1: four quadratic terms $T_1,\dots,T_4$ and three cubic terms T_5. Each term is a vertex with the listed slots; T_3, T_4 carry one derivative on the leg A. For $\xi\in S_-\otimes W^*$ the table holds with $\lambda_\pm$, $\lambda^*_\pm$ exchanged. For $\eta\in\Sigma^*$,

$$S^{(1)}(\eta)=\sum_\mu\int x^\mu\,s_\mu+\int s_0,\qquad\int s_\mu:=S^{(1)}(\xi_\mu),\quad\xi_\mu:=\rho(e_\mu)\eta\in\Sigma,\qquad s_0:=-\big(\rho(A|_\perp)\eta,\lambda^*\big),\tag{1.1}$$

(eq c2-S1S, Lemma c11-lines (ii)), with e_μ an orthonormal basis of V; for $\eta \in S_+ \otimes W^*$ the spinors ξ_μ lie in $S_- \otimes W^*$ and s_0 has the slots $(A|_\perp, \lambda^*_+)$, for $\eta \in S_- \otimes W$ they lie in $S_+ \otimes W$ and s_0 has the slots $(A|_\perp, \lambda^*_-)$. For even x with vector field v, $S^{(1)}(x) = \int \big((v.A, A^*) - (v.\lambda, \lambda^*) - (v.c, c^*)\big)$ (Def c7-vdot), quadratic, with the five slot pairs $(A|_V, A^*|_V)$, $(A|_\perp, A^*|_\perp)$, $(\lambda_\pm, \lambda^*_\pm)$, (c, c^*).

Table 1: the terms of $S^{(1)}(\xi)$, $\xi \in S_+ \otimes W$, and the intrinsic term of $S^{(1)}(\eta)$. The two-component forms are those of Rem c8-twist for ${\xi_\alpha}^i = u_\alpha z^i$, $\bar\xi = 0$, and of Def c8-twocomp for $\rho(A|_\perp)\eta$; for general ξ replace $u_\alpha z^i$ by ${\xi_\alpha}^i$.

term	index-free	two-component, ${\xi_\alpha}^i = u_\alpha z^i$ (Rem c8-twist)	slots
T_1	$-(\Gamma(\xi,\lambda)\|_V, A^*\|_V)$	$-i\big(\bar\lambda_{\dot\alpha i} z^i \bar\sigma^{m\dot\alpha\alpha} u_\alpha, A^*_m\big)$	λ_-, $A^*\|_V$
T_2	$-(\Gamma(\xi,\lambda)\|_\perp, A^*\|_\perp)$	$\frac12 \varepsilon_{jikl} z^j \big((u\lambda^i), A^{*kl}\big)$	λ_+, $A^*\|_\perp$
T_3	$\frac12(\rho(dA\|_V)\xi, \lambda^*)$	$-\frac14\big(F^0_{mn}(\sigma^m\bar\sigma^n u)_\alpha, \lambda^{*\alpha}{}_i z^i\big)$, $F^0 = dA\|_V$	$A\|_V$ (one derivative), λ^*_+
T_4	$\frac12(\rho(dA\|_\perp)\xi, \lambda^*)$	$-\frac{i}{2}\big(u^\alpha \sigma^m_{\alpha\dot\alpha}\partial_m(A_{ij}z^j), \bar\lambda^{*\dot\alpha i}\big)$	$A\|_\perp$ (one derivative), λ^*_-
T_5	$\frac12(\rho(\frac12[A\wedge A])\xi, \lambda^*)$	the commutator terms of F_{mn}, $D_m(A_{ij}z^j)$ and $-\frac14\big([A^{ik}, A_{kj}z^j]u_\alpha, \lambda^{*\alpha}{}_i\big)$	$(A\|_V, A\|_V, \lambda^*_+)$, $(A\|_V, A\|_\perp, \lambda^*_-)$, $(A\|_\perp, A\|_\perp, \lambda^*_+)$
s_0	$-(\rho(A\|_\perp)\eta, \lambda^*)$	$\big(A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \lambda^{*\alpha}{}_k\big) - \big(A_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}, \bar\lambda^{*\dot\beta k}\big)$	$(A\|_\perp, \lambda^*_\pm)$

The census of Section 4 uses only the slot structure of Table 1 and of Def c1-vertex; the engines use the explicit realisation of $S^{(1)}$ in `superconformal/model.py` (`S1_odd`, `S1_even`), which is the realisation of Chapter 2.

Definition 1.2 (Defect cycles and two-body kernels)**.** Let x be a generator, X_a, X_b two letters of any slots, $0 < \varepsilon < T \le \infty$.

- (i) A defect cycle of the pair (X_a, X_b) along x is a chain $X_a - e_0 - V_1 - e_1 - \cdots - V_n - e_n - X_b$ (Def c10-chain) whose vertices V_j are vertices of I and exactly one vertex V_s given by a term of $S^{(1)}(x)$, entered and left through two of its slots (the third slot of a cubic term is an output), and whose edges are components of $P(\varepsilon, T)$ and exactly one special edge $e_\sigma = K_\varepsilon - K_0$, with the components of K_0 (a field slot with its antifield slot, either one possibly a slot of a letter). The vertex V_s is placed in one of three ways: *on a line*, with both legs on edges P; *at the special edge*, with one leg on e_σ; *cubic*, a cubic term with its two cycle legs on edges P or one of them on e_σ.

- (ii) $\mathcal{D}_x(X_a, X_b; \varepsilon, T)$ is the sum of the one-loop graphs of the flow of the defect vertex of Lemma c8-transfer, applied to $e^{(J+\delta M(\varepsilon))/\hbar}$ with $M = \int \omega\, X_a X_b$, that are δ-linear, of arity one along x, and whose cycle is a defect cycle of the pair, amputated at the outputs, the outputs left open as in Def c10-pair. The signs are those of the flow: the edge rule of Prop c3-graphs, the signs of Lemma c10-sign along the chain from X_a, and the sign of the special edge of Prop c11-master ($\{f,g\}_\varepsilon - \{f,g\}$ is minus f and g joined by $K_\varepsilon - K_0$). Operationally, a chain read from X_a with V_s at position s and the special edge e_σ carries, beyond the sign of Lemma c10-sign, the factor $-(-1)^{n(<\sigma)+|S^{(1)}(x)|\,(n(<s)+[s<\sigma])}$, with $n(<j)$ the number of odd outputs before position j, $[s<\sigma] = 1$ if V_s precedes e_σ, and $|S^{(1)}(x)|$ the total parity of the functional $S^{(1)}(x)$: 0 for odd x, 1 for even x, for which this is the factor of Def c11-sp up to the overall sign of the special edge. In code the rule carries no free constant: route A applies, on top of the chain sign of the two-letter engine and the pair sign of Lemma c10-sign, the factor $(-1)^{n(<\sigma)+|S^{(1)}(x)|(n(<s)+[s<\sigma])}$ and a further -1 when e_σ is the edge at X_b (the edge reading of the special edge between slots of

opposite parity); route B applies $+\frac{\hbar}{2}\sum s_x\,\mathrm{STr}$ with s_x this factor without its leading minus; both reproduce G-reg, G-pole, the graded symmetry under $X_a \leftrightarrow X_b$ and G-R1 (Section 9).

- (iii) The finite special vertex of the pair is

$$\mathcal{F}_x(X_a,X_b) := -\sum_{m\geq 1}\frac{(-1)^m}{m!}\Big(\big\{I,\mathrm{ct}^{(1)}_{x,m}(X_aX_b)\big\}_{H^m}+\big\{S^{(1)}(x),\mathrm{ct}_m(X_aX_b)\big\}_{H^m}\Big), \tag{1.2}$$

  with $\{-,-\}_{H^m}$ the bracket built from $(H^m\otimes 1)K_0$ in place of K_0 (eq c10-finitesp).

- (iv) The two-body kernel of (X_a,X_b) along x is

$$\Upsilon_x(X_a,X_b) := \big[\varepsilon^0\big]\,\mathcal{D}_x(X_a,X_b;\varepsilon,T)+\mathcal{F}_x(X_a,X_b), \tag{1.3}$$

  the coefficient of ε^0 in the expansion of Theorem 5.2, a local density of the outputs at the point of the pair with values in the tensor product of the fibres and colour spaces of the two letters.

For $x\in\{\Delta,P_b\}$ and the placement on a line, $\mathcal{D}_x$ restricted to these cycles is the special-edge composite cycle Z_x of Def c11-sp.

2 The generators on the regulator

For an odd generator the operator $x.$ is the odd linear operator on $\mathscr{E}$ that induces, as in [C07, §6.2], the derivation given by the bracket with the quadratic part of $S^{(1)}(x)$ (Prop c7-susy); for an even generator it is $v.$ of Def c7-vdot. A linear operator X acts on functionals as the derivation induced by $\ell\mapsto \ell\circ X$; $[-,-]$ is the graded commutator; $g(H):=\int_\varepsilon^T e^{-tH}dt$, so that $P\star = Q^{GF}g(H)$ and $QP\star+P\star Q = e^{-\varepsilon H}-e^{-TH} =: E(H)$ (Prop c9-composite (iii)). For a constant-coefficient operator D and a coordinate x^μ, $\sigma^\mu(D):=[D,x^\mu]$ is its symbol along e_μ, a constant-coefficient operator of order one less.

Lemma 2.1 (The arrows of $\xi.$ and of Q^{GF})**.** *For $\xi\in\Sigma$, $\xi.$ maps the slot of λ to the slot of A by $\lambda\mapsto\Gamma(\xi,\lambda)$, the slot of A to the slot of λ by $A\mapsto\frac12\rho(dA)\xi$, the slot of A^* to the slot of λ^* and the slot of λ^* to the slot of A^* (the arrows forced by the preservation of $\langle\,,\rangle$), and vanishes on the slots of c and c^*. Q^{GF} maps $A^*\mapsto -2A^*$ (slot of A), $A\mapsto -d^*(A|_V)$ (slot of c), $c^*\mapsto dc^*$ (slot of $A^*|_V$), $\lambda^*\mapsto -\not{d}|_{\Sigma^*}\lambda^*$ (slot of λ), and vanishes on c and λ (Table c3, Chapter 3).*

Proof. The terms $-(\Gamma(\xi,\lambda),A^*)$ and $\frac12(\rho(dA)\xi,\lambda^*)$ of $S^{(1)}(\xi)$ pair the slots (λ,A^*) and (A,λ^*); the bracket of a quadratic functional $\int(Xe,e^*)$ with a coordinate functional is the coordinate functional composed with the corresponding arrow (proof of Lemma c9-SDelta (i)), and each pairing term gives one arrow between field slots and the transposed arrow between antifield slots (proof of Prop c7-susy). The signs: in the reading of Prop c1-Q the term (dc,A^*) of the free action eq c1-free gives $c\mapsto -dc$, so the term $-(\Gamma(\xi,\lambda),A^*)$, whose letter λ is odd like c, gives $\lambda\mapsto+\Gamma(\xi,\lambda)$; the antifield arrow $A^*\to\lambda^*$ is its transpose for $\langle\,,\rangle$, and $[\xi.,Q]=0$ on the slot of A (the part of polynomial degree two of $\{S,S^{(1)}(\xi)\}=0$) then fixes $A\mapsto+\frac12\rho(dA)\xi$. This is the classical component of Cor c7-Q, which replaces A by $\Gamma(\xi,\lambda)$ and gives $\{S^{(1)}(\xi),\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)\}=+F_1$ (the first functional listed there), and both engine realisations of $S^{(1)}(\xi)$ confirm it (task E0: the field arrows read off the vertex blocks are $+2$ times these, the factor 2 being the vertex normalisation of route A, and equal to them in route B). The table of Q^{GF} is Chapter 3. □

Proposition 2.2 (Commutators with the regulator)**.** • *(i) Supercharges. For $\xi\in\Sigma$: $[\xi.,Q]=0$, $[\xi.,H]=0$, and $\xi.$ preserves $\langle\,,\rangle$. The even operator*

$$C_\xi := [\xi.,Q^{GF}] = \xi.\,Q^{GF}+Q^{GF}\xi. \tag{2.1}$$

of degree -1 has constant coefficients, every component of order exactly one, and the four components

$$\lambda \mapsto -d^*\big(\Gamma(\xi,\lambda)|_V\big)\ (\textit{slot of } c), \qquad A^*|_V \mapsto \lambda, \qquad \lambda^* \mapsto A|_V, \qquad c^* \mapsto \lambda^*,$$

each nonzero for $\xi \neq 0$; it vanishes on the slots of A, c and $A^*|_\perp$, and has no component in the slot of $A|_\perp$. For $\xi \in S_+ \otimes W$ the four components are $\lambda_- \to c$, $A^*|_V \to \lambda_+$, $\lambda_+^* \to A|_V$, $c^* \to \lambda_-^*$ (for $\xi \in S_- \otimes W^*$ exchange the chiralities). On a plane wave $e^{i(p,x)}$ the first is $\lambda \mapsto +i\big(p, \Gamma(\xi,\lambda)|_V\big)$, and the second is a nonzero multiple of $A^*|_V \mapsto (p, A^*|_V)\,\xi$.

- *(ii) Special supercharges. For $\eta \in \Sigma^*$, with $\xi_\mu = \rho(e_\mu)\eta$, $\eta^0.$ the constant operator of order zero induced by s_0 of (1.1) (arrows $A|_\perp \to \lambda$ and $\lambda^* \to A^*|_\perp$), and Y_η the constant operator of order zero from λ^* to A^* produced when the derivative of the terms T_3, T_4 of $\int x^\mu S_2(\xi_\mu)$ is moved off A in the transposed (antifield) arrow, $\partial_\nu(x^\mu\,\cdot) = x^\mu\partial_\nu + \delta^\mu_\nu$,*

$$\begin{aligned} &\eta. = \sum_\mu x^\mu\,\xi_\mu. + \eta^0. + Y_\eta, \qquad [\eta., Q] = \sum_\mu \sigma^\mu(Q)\,\xi_\mu. + [\eta^0. + Y_\eta, Q] = 0,\\ &[\eta., H] = 2\sum_\mu \xi_\mu.\,\partial_\mu, \end{aligned} \tag{2.2}$$

$$[\eta., Q^{GF}] = \sum_\mu x^\mu\, C_{\xi_\mu} + C^0_\eta, \qquad C^0_\eta := \sum_\mu \sigma^\mu(Q^{GF})\,\xi_\mu. + [\eta^0. + Y_\eta, Q^{GF}],$$

with C^0_η constant-coefficient of order at most one. Without Y_η the second identity fails. In particular $\eta.$ commutes with neither H nor Q^{GF}, and the second identity of (2.2) is the conformal Killing equation of $\rho(x)\eta$ in operator form.

- *(iii) Conformal Killing fields. For an even generator with conformal Killing field v, $\partial_\mu v_\nu = \frac14(\operatorname{div} v)\delta_{\mu\nu} + \omega_{\mu\nu}$ with ω antisymmetric, write $v. = v^\mu\partial_\mu + \frac12\omega_{\mu\nu}\Sigma^{\mu\nu} + \frac{w}{4}\operatorname{div} v$ on a slot of Weyl weight w, with $\Sigma^{\mu\nu}$ the rotation generators of the slot ($(\Sigma^{\mu\nu}A)_m = \delta_{m\mu}A_\nu - \delta_{m\nu}A_\mu$ on one-forms). Then $[v., Q] = 0$, and for every rotation-covariant constant-coefficient operator D of weight -2 (an operator of order k mapping a slot of weight w to one of weight $w+k-2$), in particular $D = Q^{GF}$,*

$$[v., D] = -\tfrac12(\operatorname{div} v)\,D - \tfrac{w}{4}\,\sigma^\rho(D)\,\partial_\rho(\operatorname{div} v) - \tfrac12\,\sigma^\rho(D)\,(\partial_\rho\omega_{\mu\nu})\,\Sigma^{\mu\nu}, \tag{2.3}$$

$$[v., H] = -\tfrac12(\operatorname{div} v)\,H + \tfrac{w-1}{2}\,(\partial_\rho \operatorname{div} v)\,\partial_\rho + (\partial_\rho\omega_{\mu\nu})\,\Sigma^{\mu\nu}\partial_\rho. \tag{2.4}$$

For $v = 2E$ these are $[v., Q^{GF}] = -4Q^{GF}$ and $[v., H] = -4H$ (Prop c7-dil); for K_b, $\operatorname{div} v = -8(b,x)$, $\omega_{\mu\nu} = 2(x_\mu b_\nu - x_\nu b_\mu)$, and

$$\begin{aligned} &[v., Q^{GF}] = 4(b,x)\,Q^{GF} + C_K, \qquad C_K := 2\,\sigma^\rho(Q^{GF})\big(w\,b_\rho - b_\nu\Sigma^{\rho\nu}\big),\\ &[v., H] = 4(b,x)\,H - 4(w-1)(b,\partial) + 4\,b_\nu\Sigma^{\rho\nu}\partial_\rho, \end{aligned} \tag{2.5}$$

C_K of order zero with constant coefficients; on the slot of $A^*|_\perp$, $C_K = 0$ and $[v., Q^{GF}]$ is multiplication by $\operatorname{div} v$ (eq c7-divv); on the slot of $A|_V$, $C_K A|_V = -4(b, A|_V)$ in the slot of c.

- *(iv) For translations, $\mathfrak{so}(V)$ and $\mathfrak{so}(W^\perp_{\mathbb{R}})$ all these commutators vanish (Prop c7-cov).*

Proof. (i) $[\xi., Q] = 0$, $[\xi., H] = 0$ and the preservation of the pairing are Prop c7-susy (i). By Lemma 2.1, $\xi.Q^{GF}$ has the arrows $A^* \to A \to \lambda$, $c^* \to A^*|_V \to \lambda^*$, $\lambda^* \to \lambda \to A$, and $Q^{GF}\xi.$ the arrows $\lambda \to A \to c$, $A^* \to \lambda^* \to \lambda$, $\lambda^* \to A^* \to A$; the arrows from A and from c end on slots where the second factor vanishes. On λ, $C_\xi\lambda = Q^{GF}(\Gamma(\xi,\lambda)) = -d^*(\Gamma(\xi,\lambda)|_V)$, and $d^* = -\iota_\partial$ becomes $-i\iota_p$. On a plane wave $A^* = a\,e^{i(p,x)}$ the two composites from A^* to λ are opposite multiples of $\rho(p)\rho(a)\xi$ plus,

from $\rho(q(p\wedge a))=\rho(p)\rho(a)-(p,a)$ [ESW, (29)] in the first, a multiple of $(p,a)\,\xi$; for $a\in W_{\mathbb{R}}^{\perp}$, $p\wedge a$ has no contraction and the two cancel, so the component is $A^*|_V\mapsto$ a nonzero multiple of $(p,a)\,\xi$; its transpose $\lambda^*\to A$ lands in $A|_V$. Route B (task E0) finds exactly the four blocks $\lambda_-\to c$, $A^*|_V\to\lambda_+$, $\lambda_+^*\to A|_V$, $c^*\to\lambda_-^*$ for $\xi\in S_+\otimes W$, and route A the same restriction to V. Each arrow of $\xi.$ and of Q^{GF} has order zero or one and the composites have order one: Q^{GF} has order one from A and from c^*, λ^* and order zero from A^*, $\xi.$ order one from A and from λ^* and order zero from λ and A^*. Nonvanishing: on λ the component is nonzero because $\Gamma(\xi,-)|_V$ is nonzero on one chirality of λ for $\xi\neq 0$; the component $A^*\to\lambda$ is nonzero by Prop c7-susy (ii); the components $\lambda^*\to A$ and $c^*\to\lambda^*$ are the transposes of $A^*\to\lambda$ and $\lambda\to c$ for the pairing, because $\xi.$ preserves the pairing and Q^{GF} has the adjointness of Prop c3-adjoint, so that $(C_\xi\otimes 1)K_0$ is graded symmetric.

(ii) By (1.1) the quadratic part of $S^{(1)}(\eta)$ is $\sum_\mu\int x^\mu\,S_2(\xi_\mu)+\int s_0$, with S_2 the quadratic part; the field arrows of $\int x^\mu S_2(\xi_\mu)$ are x^μ times those of $\xi_\mu.$, and its antifield arrows, the transposes for $\langle\,,\rangle$, are obtained by integrating by parts the derivative of T_3, T_4 against $x^\mu\lambda^*$, which gives x^μ times the antifield arrows of $\xi_\mu.$ plus the constant arrows Y_η from λ^* to A^*; this gives the decomposition of $\eta.$ (route B, task E0, checks $[\eta.,Q]=0$ with Y_η and its failure without). For odd Q and even x^μ, $[x^\mu\xi_\mu.,Q]=x^\mu[\xi_\mu.,Q]+[Q,x^\mu]\xi_\mu.$, and $[\xi_\mu.,Q]=0$ by (i). The vanishing of $[\eta.,Q]$ is the part of polynomial degree two of the arity-one equation $\{S,S^{(1)}(\eta)\}=0$ (Lemma c2-components), as in Prop c7-dil. H is $-\partial^2$ on every component and $[x^\mu,-\partial^2]=2\partial_\mu$; $\eta^0.$ and $\xi_\mu.$ have constant coefficients and commute with H by (i). The formula for $[\eta.,Q^{GF}]$ is the same computation with Q^{GF} in place of Q.

(iii) $[v.,Q]=0$ is Prop c7-dil. For the transport term, $[v^\mu\partial_\mu,D]=-\sigma^\nu(D)(\partial_\nu v^\mu)\partial_\mu=-\frac14(\operatorname{div}v)D^{(1)}-\sigma^\nu(D)\omega_\nu{}^\mu\partial_\mu$, with $D^{(1)}$ the part of order one; for the spin term, $[\frac12\omega_{\mu\nu}\Sigma^{\mu\nu},D]=\frac12\omega_{\mu\nu}(\Sigma'^{\mu\nu}D-D\Sigma^{\mu\nu})-\frac12\sigma^\rho(D)(\partial_\rho\omega_{\mu\nu})\Sigma^{\mu\nu}$, with Σ' the generators of the target slot; rotation covariance of D cancels $-\sigma^\nu(D)\omega_\nu{}^\mu\partial_\mu+\frac12\omega_{\mu\nu}(\Sigma'^{\mu\nu}D-D\Sigma^{\mu\nu})$ pointwise; for the weight term, $\frac{w'}{4}(\operatorname{div}v)D-D\frac{w}{4}(\operatorname{div}v)=\frac{w'-w}{4}(\operatorname{div}v)D-\frac{w}{4}\sigma^\rho(D)\partial_\rho(\operatorname{div}v)$, and $w'-w=k-2$ gives $-\frac14-\frac14=-\frac12$ on the part of order one and $-\frac12$ on the part of order zero. For $H=-\partial^2$: $[v^\mu\partial_\mu,-\partial^2]=2(\partial_\rho v^\mu)\partial_\rho\partial_\mu+(\partial^2v^\mu)\partial_\mu=-\frac12(\operatorname{div}v)H+(\partial^2v^\mu)\partial_\mu$, and $\partial^2v_\mu=-\frac12\partial_\mu\operatorname{div}v$ for a conformal Killing field of $\mathbb{R}^4$; $[f,-\partial^2]=2(\partial_\rho f)\partial_\rho+\partial^2 f$ for a multiplication operator, $\partial^2\omega=0$ and $\partial^2\operatorname{div}v=0$. For K_b, $\partial_\mu v_\nu=2x_\mu b_\nu-2b_\mu x_\nu-2(b,x)\delta_{\mu\nu}$ and $\partial_\rho\omega_{\mu\nu}=2(\delta_{\rho\mu}b_\nu-\delta_{\rho\nu}b_\mu)$, which gives (2.5). On $A^*|_\perp$, $Q^{GF}=-2$ has order zero and $\sigma^\rho(Q^{GF})=0$; on $A|_V$, $Q^{GF}A|_V=\partial_\rho A_\rho$, $\sigma^\rho(Q^{GF})Y=Y_\rho$, $w=1$ and $\sigma^\rho(Q^{GF})\Sigma^{\rho\nu}A=3A_\nu$, so $C_KA|_V=2(b,A|_V)-6(b,A|_V)$. (iv) Prop c7-cov. □

Corollary 2.3 (Variation kernels and composite lines)**.** *Let $P=P(\varepsilon,T)$.*

- *(i) For every generator, $(x.\otimes 1+1\otimes x.)K_t=([x.,e^{-tH}]\otimes 1)K_0$, and the variation kernel of the propagator is*

$$\mathcal{V}_x(\varepsilon,T):=(x.\otimes 1+1\otimes x.)P(\varepsilon,T)=\int_\varepsilon^T\Big(\big([x.,Q^{GF}]\,e^{-tH}+Q^{GF}[x.,e^{-tH}]\big)\otimes 1\Big)K_0\,dt.$$

  Along $\xi\in\Sigma$, $\mathcal{V}_\xi=(C_\xi\,g(H)\otimes 1)K_0$, graded symmetric, with components in $\lambda\otimes A|_V$, $A|_V\otimes\lambda$, $c\otimes\lambda^*$, $\lambda^*\otimes c$ (none in the slots of $A|_\perp$), of momentum degree -1. Along η, $[\eta.,e^{-tH}]=-2t\sum_\mu\xi_\mu.\partial_\mu e^{-tH}$ and $\mathcal{V}_\eta=\int_\varepsilon^T\big((\sum_\mu x^\mu C_{\xi_\mu}+C^0_\eta)e^{-tH}-2tQ^{GF}\sum_\mu\xi_\mu.\partial_\mu e^{-tH}\big)\otimes 1\big)K_0\,dt$. Along Δ, $\mathcal{V}_\Delta=-4(\varepsilon\partial_\varepsilon+T\partial_T)P$ (Prop c9-ward). Along K_b, the parts $4(b,x)Q^{GF}$, $4(b,x)H$ of (2.5) give the endpoint terms $\frac18\operatorname{div}v(m)\,(4\varepsilon L_\varepsilon-4TL_T)$, m the midpoint of the two arguments, and C_K, $-4(w-1)(b,\partial)$, $4b_\nu\Sigma^{\rho\nu}\partial_\rho$ give bulk terms linear in the separation, present at every T (k_transfer, Lemma ktr-noninv, which states them component by component on $A\otimes A$, $A|_\perp\otimes A|_\perp$ and $c\otimes A^*_m$).

- *(ii) For every generator whose operator $x.$ preserves the pairing, the composite kernel along x, the two legs of the quadratic part of $S^{(1)}(x)$ joined in both orders to two edges P, is*

$$\big(P\star x.\otimes 1+1\otimes P\star x.\big)P=2\big(P\star x.\,P\star\otimes 1\big)K_0.$$

Its components lie in $c\otimes\lambda$ and $\lambda\otimes c$ for odd x (the term $-(\Gamma(\epsilon(y),\lambda)|_V, A^*|_V)$ between a $\lambda\otimes\lambda$ and an $A^*|_V\otimes c$ component, ϵ the Killing spinor of x), and in $c\otimes A|_V$, $A|_V\otimes c$ for even x (Lemma c11-kernels (iii), (iv)).

- *(iii) For $\xi\in\Sigma$,*

$$P\star\,\xi.\,P\star = C_\xi\,Q^{GF}g(H)^2 - \xi.\,(Q^{GF})^2 g(H)^2, \qquad \big[Q,\,P\star\,\xi.\,P\star\big] = E(H)\,C_\xi\,g(H)\neq 0,$$

both terms of the first identity being nonzero ($C_\xi\neq 0$, $(Q^{GF})^2\neq 0$). Hence $(Q\otimes 1+1\otimes Q)$ maps the composite kernel along ξ to $2(E(H)C_\xi g(H)\otimes 1)K_0$, the variation kernel $\mathcal{V}_\xi$ with the factor $e^{-\varepsilon|p|^2}-e^{-T|p|^2}$ in place of 1: unlike along Δ and P_b (Lemma c11-kernels (v)), the composite kernel along ξ is not annihilated by $Q\otimes 1+1\otimes Q$. Along η and K_b the same holds, with the additional terms of (ii), (iii) of Proposition 2.2.

- *(iv) Weight rule. Every component of $\mathcal{V}_x$ and of the composite kernel along x between slots of Weyl weights w_1, w_2 has momentum degree $w_1+w_2-4-\frac12\lambda_x$ once the Killing data are given the weights of Prop c7-select ($-\frac12$ to a constant spinor, -1 to a constant vector field) and the position the weight -1; the special edge has momentum degree $w_1+w_2-4=0$.*

Proof. (i) As in Prop c7-cov and Prop c7-susy, preservation of the pairing gives $(x.\otimes 1+1\otimes x.)K_0=0$ and $K_t=(e^{-tH}\otimes 1)K_0$. Along ξ, $[\xi., e^{-tH}]=0$. Along η, $[\eta., H]$ commutes with H, so $[\eta., e^{-tH}] = -t[\eta., H]e^{-tH}$. Along K_b, write $v.=-(b,x)\,v_\Delta.+(\text{rest})$ pointwise; the part of (2.5) proportional to (b,x) is $-(b,x)$ times the dilatation commutators, which integrate to the endpoint terms of Prop c9-ward with the factor $\frac18\operatorname{div} v$ evaluated at the midpoint by the symmetrisation over the two arguments; the remaining terms are of order zero or one with constant coefficients times the separation (k_transfer Lemma ktr-noninv, verified symbolically there). (ii) The computation of Prop c9-composite (i) with the Koszul signs of Lemma c9-ops: for odd $x.$, $P\star x.$ is even, $(1\otimes P\star x.)(P\star\otimes 1)K_0 = (P\star\otimes 1)(1\otimes P\star)(1\otimes x.)K_0$, $(1\otimes x.)K_0=-(x.\otimes 1)K_0$, $(1\otimes P\star)(x.\otimes 1)=-(x.\otimes 1)(1\otimes P\star)$ and $(1\otimes P\star)K_0=(P\star\otimes 1)K_0$, so both orders give $(P\star x.P\star\otimes 1)K_0$. The components follow from Lemma 2.1: $\lambda^*\to\lambda\to A\to c$ and $c^*\to A^*|_V\to\lambda^*\to\lambda$; the arrows of $P\star$ from A^* and of $\xi.$ from A end on slots where Q^{GF} vanishes. (iii) $[\xi., P\star]=C_\xi g(H)$ since $[\xi., g(H)]=0$; then $P\star\xi.P\star=(C_\xi g-\xi.P\star)P\star$. With $E=QP\star+P\star Q$ and $[\xi., Q]=0$, $Q(P\star\xi.P\star)=E\xi.P\star-P\star Q\xi.P\star=E\xi.P\star+P\star\xi.E-P\star\xi.P\star Q$, so $[Q,P\star\xi.P\star]=E(\xi.P\star+P\star\xi.)=EC_\xi g$, E commuting with $\xi.$. (iv) The count of Lemma c10-degree with the vertex $S^{(1)}(x)$ of weight $4-\frac12\lambda_x$ on its fields. □

Proposition 2.4 (Ward identity of the flow along ξ)**.** *Let $\xi\in\Sigma$, $\xi.F$ the derivation of functionals induced by $\xi.$, $P=P(\varepsilon,T)$, J even and $W=W(P,J)$. Then*

$$\xi.\,W = e^{-W/\hbar}e^{\hbar\partial_P}\big((\xi.J)\,e^{J/\hbar}\big) - \hbar\,\partial_{\mathcal{V}_\xi}W - \tfrac12\big(\partial_{\mathcal{V}_\xi}(W^2) - 2W\,\partial_{\mathcal{V}_\xi}W\big). \tag{2.6}$$

The first term is the linearised flow of $\xi.J$, the second the graphs of W with one edge replaced by $\mathcal{V}_\xi$ on a cycle or a tree, the third two copies of W joined by $\mathcal{V}_\xi$. For $J=I+\delta M$, $\xi.I=\pm\big(QS_3(\xi)+\{I,S_3(\xi)\}\big)$ by the arity-one equation $QS^{(1)}(\xi)+\{I,S^{(1)}(\xi)\}=0$, S_3 the cubic part, with the sign of the bracket conventions.

Proof. By Lemma c9-ops, $[\partial_P,\xi.]=\partial_{\mathcal{V}_\xi}$, and $\partial_{\mathcal{V}_\xi}$ commutes with ∂_P (constant-coefficient second-order operators). Hence $\partial_P^n\xi. = \xi.\partial_P^n + n\,\partial_{\mathcal{V}_\xi}\partial_P^{n-1}$ and $\xi.\,e^{\hbar\partial_P} = e^{\hbar\partial_P}\xi. - \hbar\,\partial_{\mathcal{V}_\xi}e^{\hbar\partial_P}$. Apply to $e^{J/\hbar}$, use $\xi.e^{F/\hbar}=\hbar^{-1}(\xi.F)e^{F/\hbar}$ for the derivation $\xi.$ and the product rule of the odd second-order operator $\partial_{\mathcal{V}_\xi}$ (proof of Prop c9-conj), and multiply by $\hbar\,e^{-W/\hbar}$. The last statement is the part of polynomial degree at least three of $\{S,S^{(1)}(\xi)\}=0$. □

Remark 2.5 (What the Ward identity shows)**.** Along Δ the variation kernel is a total derivative in the endpoints, $\mathcal{V}_\Delta = -4(\varepsilon\partial_\varepsilon + T\partial_T)P$, so the graphs with one variation edge collapse to the scaling of the flow and produce the term $4\,\mathrm{lc}$ of Thm c10-D1; along the translations, rotations and the R-symmetry $\mathcal{V}_x = 0$. Along ξ, $\mathcal{V}_\xi = (C_\xi g(H)\otimes 1)K_0$ is a new edge of the same infrared order as a propagator, joining λ to $A|_V$ and c to λ^*, and no identity reduces the graphs with one variation edge to a scaling. The content of Q_1 beyond the classical transformation is therefore carried by the defect cycles of Definition 1.2 themselves, which the next section isolates.

Proposition 2.6 (Position decomposition of the Killing data)**.** *Let x_O be the point of the density of a monomial O.*

- *(i) Along $\eta \in \Sigma^*$, graph by graph, the position part along b (Def c11-canonical) of a graph with the vertex $S^{(1)}(\eta)$ is the same graph with $S^{(1)}(\rho(b)\eta)$, $\rho(b)\eta \in \Sigma$; its translation-invariant part is the graph with $S^{(1)}(\eta)$ at the Killing spinor $\rho(x - x_O)\eta$ together with $\int s_0$ (Prop c11-S (i)). On plane waves, with a momentum r injected at the vertex and absorbed by ω, the factor $(x - x_O)^\mu$ is $-i\,\partial/\partial r_\mu$ acting on the integrand at $r = 0$.*

- *(ii) Along K_b, with $z = x - x_O$,*

$$v(x) = v(x_O) + \big(-2(b, x_O)\, z + \omega_{x_O} z\big) + v(z), \qquad \omega_{x_O} z := 2\big((x_O, z)\, b - (b, z)\, x_O\big),$$

  so that, $S^{(1)}$ being linear in the Killing data with its weight terms,

$$S^{(1)}(K_b) = S^{(1)}\big(P_{v(x_O)}\big) - (b, x_O)\, S^{(1)}\big(\Delta_{x_O}\big) + S^{(1)}\big(M_{\omega_{x_O}, x_O}\big) + S^{(1)}\big(K_{b,x_O}\big), \tag{2.7}$$

  where the subscript x_O denotes the generator centred at x_O (vector fields $2z$, $\omega_{x_O} z$, $v(z)$). Graph by graph the corresponding parts are the graphs along the translation $P_{v(x_O)}$, $-(b, x_O)$ times the translation-invariant part along Δ, the translation-invariant part along the rotation $M_{\omega_{x_O}}$, and the intrinsic part along K_b, computed with the factor $z^\mu z^\nu$ realised as $-\partial^2/\partial r_\mu \partial r_\nu$ at $r = 0$.

- *(iii) Hence, for the two-body kernels at the point x_O of the pair,*

$$\Upsilon_\eta = \sum_\mu x_O^\mu\, \Upsilon_{\xi_\mu} + \Upsilon^0_\eta, \qquad \Upsilon_{K_b} = v(x_O)^\mu\, \Upsilon_{P_{e_\mu}} - (b, x_O)\, \Upsilon^0_\Delta + \Upsilon^0_{M_{\omega x_O}} + \Upsilon^0_{K_b},$$

  with Υ^0_y the translation-invariant (intrinsic) part, computed with the generator centred at the point of the pair; Υ^0_η has Weyl weight $w_{X_a} + w_{X_b} - \frac{1}{2}$ and $\Upsilon^0_{K_b}$ Weyl weight $w_{X_a} + w_{X_b} - 1$, both translation invariant.

Proof. (i) Prop c11-S (i), whose argument (Prop c11-position (i)) applies to the special edges, to the vertices $S^{(1)}(\eta)$ on the cycle in each of the three placements, and to the counterterm vertices. (ii) Direct expansion of $(x_O + z, x_O + z)b - 2(b, x_O + z)(x_O + z)$; the divergences add, $\operatorname{div} v(x) = -8(b, x_O) - 8(b, z)$, with $\operatorname{div}(2z) = 8$ and $\operatorname{div}(\omega z) = 0$, and $S^{(1)}$ of Def c7-vdot is linear in v including the weight terms $\frac{w}{4}\operatorname{div} v$ and the spin terms. In a graph the vertex at $y = x_O + z$ carries z polynomially; the argument of (i) applies to each monomial in z, the second-order monomials giving second derivatives in r. (iii) The kernels are linear in the generator; insert (i), (ii) into Definition 1.2, the finite special vertices decomposing in the same way. The weights: Prop c7-select. □

3 The defect graphs and the two-body form of the components

Let x be a generator and $O = \int \omega\, X_1 \cdots X_n$ a monomial local functional of degree $|O|$; sums of monomials are treated by linearity. $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - \hbar F_1$ with $\hbar F_1$ a one-loop quantisation

through arity one along x: the equivariant master equation eq c8-qme modulo $\hbar^2$ holds in arity zero and in arity one along x. The finite part of the operators is $\psi = 0$; its effect is the law (b) of Prop c8-laws (Section 6). The endpoint is $T = \infty$ for $x \in \Sigma \cup \Sigma^*$ (Prop c8-Tsafe) and any $T \in (0, \infty)$ for $x = K_b$.

Proposition 3.1 (The defect vertex at arity one)**.** *By Lemma c8-transfer, $d_T\, O[T]$ is the $\varepsilon \to 0$ limit of the flow, with edges $P(\varepsilon, T)$, of the vertex $(d_{CE} + Q)\mathcal{J} + \frac{1}{2}\{\mathcal{J}, \mathcal{J}\}_\varepsilon + \hbar\Delta_\varepsilon \mathcal{J}$, $\mathcal{J} = J + \delta\, O(\varepsilon)$. Its part linear in δ, of arity one along x and of order at most $\hbar$, is the sum of:*

- *(V1) the classical vertex $\delta\,(sO)^{(1)}(x) = \pm\delta\{S^{(1)}(x), O\}$, and the arity-zero vertex $\delta\,(Q + \{I, -\})O$ joined to a separate vertex $S^{(1)}(x)$;*
- *(V2) the special-edge vertices of order $\hbar^0$: $\delta\big(\{S^{(1)}(x), O\}_\varepsilon - \{S^{(1)}(x), O\}\big)$ (special edge between $S^{(1)}(x)$ and O), $\delta\big(\{I, O\}_\varepsilon - \{I, O\}\big)$ with a separate $S^{(1)}(x)$ (between I and O), $\{I, S^{(1)}(x)\}_\varepsilon - \{I, S^{(1)}(x)\}$ with a separate O (between I and $S^{(1)}(x)$), $\frac{1}{2}(\{I, I\}_\varepsilon - \{I, I\})$ with separate O and $S^{(1)}(x)$ (between two vertices I), and the self-contraction $\delta\,\hbar\Delta_\varepsilon O$ through K_ε with a separate $S^{(1)}(x)$;*
- *(V3) the counterterm vertices of order $\hbar$: $-\delta\big(Q\,\mathrm{ct}_x^{(1)}(O) + \{I, \mathrm{ct}_x^{(1)}(O)\}_\varepsilon + \{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon + \{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x) + \hbar F_1^{(1)}(x), O\}_\varepsilon\big)$; the arity-zero vertices $-\delta\big(Q\,\partial_\delta I^{CT}[\varepsilon](O) + \{I, \partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon + \{I^{CT}[\varepsilon] + \hbar F_1^{(0)}, O\}_\varepsilon\big)$ joined to a separate $S^{(1)}(x)$; and the arity-one and arity-zero defect vertices of the action, of order $\hbar$, joined to O.*

The signs are those of the defect vertex; for odd O those of the proof of Lemma c6-bvexact.

Proof. Expand the defect vertex in δ, in arity and in $\hbar$, with $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ and $O(\varepsilon) = O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)$, whose arity-one part is $-\mathrm{ct}_x^{(1)}(O)$ (Def c8-renorm). At order $\hbar^0$ the classical master equation $(d_{CE} + Q)I_{\mathfrak{psl}(4|4)} + \frac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\} = 0$ (eq c2-mc) removes the δ-independent bracket parts and leaves their special-edge parts; the δ-linear order-$\hbar^0$ part is $(d_{CE} + Q)O + \{I_{\mathfrak{psl}(4|4)}, O\}_\varepsilon$, whose arity-one part is (V1) plus the first special edge of (V2) and whose arity-zero part, joined to $S^{(1)}(x)$ elsewhere in the graph, is the second part of (V1) and the second special edge. $\hbar\Delta_\varepsilon$ of $S^{(1)}(x)$ vanishes: its quadratic terms pair a field with the antifield of another field (Table 1, Def c7-vdot gives (f, f^*) for even x, whose K_ε self-contraction is a constant), and its cubic terms carry one antifield, λ^*, and no λ. The order-$\hbar$ terms are (V3); $\hbar\Delta_\varepsilon$ of the counterterms is of order $\hbar^2$. This is eq o-defect of the odd design with the ψ-terms and the vertex $(Q + \{I, -\})O$ included (audit findings G3, G7). □

Definition 3.2 (Counterterm cycles along x)**.** The counterterm cycles of a pair (X_a, X_b) along x are the chains of Def c10-chain with exactly one vertex given by a term of $S^{(1)}(x)$ placed on the cycle (both cycle legs on edges P; along ξ: T_1 on a line or a cubic term with its λ^* leg as output) and no special edge. $\mathrm{ct}_x^{(1)}(X_a X_b)$ is $\hbar\,\mathrm{Sing}_\varepsilon$ of their values (Def c4-ict); the arity-one counterterm of O is $\mathrm{ct}_x^{(1)}(O) = \sum_{a<b} \pm \int \omega \cdots \mathrm{ct}_x^{(1)}(X_a X_b) \cdots$ with the Koszul signs of (3.1). Its outputs carry the pair degree minus one; along Δ and P_b it is that of Def c11-graphs.

Theorem 3.3 (Two-body form of the components along every generator)**.** *Under the hypotheses above:*

- *(i) The δ-linear, arity-one, order-$\hbar$ graphs of the flow of the defect vertex fall into the classes*
  - *(F) the one-loop flow graphs of the vertices (V1);*
  - *(C2) the defect cycles of the pairs (X_a, X_b) of O (Definition 1.2) with trees of I on their outputs and the spectators;*

– *(C0) the graphs whose cycle passes through two letters of O and carries the special edge or the K_ε contraction of (V2) but no vertex $S^{(1)}(x)$, with $S^{(1)}(x)$ on a tree;*

– *(Br) the graphs in which the special edge does not lie on the cycle;*

– *(Ac) the graphs whose cycle passes through no letter of O, joined to O by a tree;*

– *(Ct) the trees with one vertex (V3).*

- *(ii) (F) is the set of one-loop graphs of the arity-one part of $(sO)[T]$ on the left side of eq c8-rep; (C0), with the arity-zero vertices of (Ct) joined to $S^{(1)}(x)$, tends to the trees of $\hbar L_1(O)^{(0)}$ with one vertex $S^{(1)}(x)$, also on the left side; (Br) tends to 0 together with the special-edge parts of the counterterm brackets (each pair is $e^{-\varepsilon|p|^2}-1$ at a generic momentum times a renormalised subgraph); (Ac), with the action defect vertices of (Ct), tends to 0 by the master equation in arity zero and in arity one along x.*

- *(iii) (C2) and the arity-one vertices of (Ct), together with the counterterm trees of the left side, the trees of $-\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x),O\})$ and of $-\mathrm{ct}^{(1)}_x(sO)$, have the limit*

$$\hbar L_1(O)^{(1)}(x) = \sum_{1\le a<b\le n} (-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1\cdots X_{a-1}\,\Upsilon_x(X_a,X_b)\,X_{a+1}\cdots\widehat{X_b}\cdots X_n \\ -\big\{\hbar F_1^{(1)}(x),O\big\}, \tag{3.1}$$

the two-body kernel Υ_x of Definition 1.2 (iv) summed over the unordered pairs of letters, and a one-body term, the bracket with the arity-one finite counterterm along x, a derivation of the letters.

- *(iv) The coefficients of $\log\varepsilon$ and of ε^{-m} cancel. Since the defect cycles contain no $\log\varepsilon$ (Theorem 5.2), the coefficient of $\log\varepsilon$ gives the arity-one logarithmic covariance along x:*

$$\mathrm{lc}\big(\{S^{(1)}(x),O\}\big)-\big\{S^{(1)}(x),\mathrm{lc}(O)\big\} = s\,\mathrm{lc}^{(1)}_x(O)-\mathrm{lc}^{(1)}_x(sO)+\big\{L^{(1)}(x),O\big\}, \tag{3.2}$$

with $L^{(1)}(x)$ the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$; and for $m\ge 1$, with the Koszul signs of (3.1),

$$\sum_{a<b}\pm\Big([\varepsilon^{-m}]\mathcal{D}_x(X_a,X_b)-\sum_{j\ge1}\tfrac{(-1)^j}{j!}\Big(\big\{I,\mathrm{ct}^{(1)}_{x,m+j}(X_aX_b)\big\}_{H^j}+\big\{S^{(1)}(x),\mathrm{ct}_{m+j}(X_aX_b)\big\}_{H^j}\Big)\Big) \\ = s\,\mathrm{ct}^{(1)}_{x,m}(O)+\{S^{(1)}(x),\mathrm{ct}_m(O)\}-\mathrm{ct}_m(\{S^{(1)}(x),O\})-\mathrm{ct}^{(1)}_{x,m}(sO), \tag{3.3}$$

the pole identity. $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ has no term ε^{-m} along Σ, Σ^*, Δ, P_b and K_b. Read for a single pair, $O=\int\omega\,X_aX_b$ with gauge-variant letters, the right side of (3.3) contains, besides the pair terms, the vacuum poles of the pairs of the cubic monomials of $\{S^{(1)}(x),O\}$ and of sO times their third letter: ct_m of the empty leg set of a pair (A,A) (Thm c10-decomp (iv)), which enters through the cubic part $\frac12(\rho(\frac12[A\wedge A])\xi,\lambda^*)$ of $S^{(1)}(\xi)$ acting on a letter λ, and $\mathrm{ct}^{(1)}_{x,m}$ of the empty leg set of a pair (c,λ), which enters through the cubic part of s on the letters; these vacuum-pole terms are needed for the identity pair by pair (route B, gate G-pole).

- *(v) (3.1) holds for every local functional, gauge-variant ones included; the operator sO enters only through the counterterm identity (3.3) and the arity-zero representative. Along K_b it holds at every $T\in(0,\infty)$ with the same right side.*

Proof. (i) At order $\hbar$ a graph of the flow has either one cycle and vertices of order $\hbar^0$, or no cycle and one vertex of order $\hbar$ (Prop c3-graphs). The arity-one, δ-linear part places x on exactly one vertex and δ on exactly one vertex. A graph with a cycle contains exactly one vertex of (V1) or one special edge (or K_ε contraction) of (V2), the other vertices being I, $S^{(1)}(x)$ (once) and O (once). With a vertex of (V1) it is in (F). With a special edge of (V2): if the cycle passes through O, it uses two letters of O (Thm c10-decomp (i)); if $S^{(1)}(x)$ and the special edge both lie on it, the graph is in (C2); if the special edge lies on it and $S^{(1)}(x)$ does not, (C0); if the special edge is not on the cycle, (Br); if the cycle avoids O, (Ac) or (Br). Trees with one vertex of order $\hbar$ are (Ct).

(ii) The left side of eq c8-rep at arity one along x is $(L(O))[T]^{(1)}(x) = \big(L(O)^{(1)}(x)\big)[T] + \big(L(O)^{(0)}\big)[T]^{(1)}(x)$, the second term being the flow of the arity-zero value with one vertex $S^{(1)}(x)$ (Thm c8-L (iv), the cochain $M[T]$ of Def c8-renorm being $C^\bullet$-linear), and $L(O) = sO + \hbar L_1(O)$. The one-loop part of $(sO)[T]^{(1)}(x)$ consists of the one-loop graphs of the vertex $(sO)^{(1)}(x) = \pm\{S^{(1)}(x), O\}$ and of the arity-zero vertex $(sO)^{(0)} = (Q + \{I, -\})O$ with one $S^{(1)}(x)$, renormalised by their counterterms: these are the graphs (F), and their counterterm trees are the two trees named in (iii) together with the arity-zero counterterms of sO joined to $S^{(1)}(x)$. For (C0): the cycle with the special edge through two letters, without $S^{(1)}(x)$, is an arity-zero defect cycle; at arity zero the same analysis (Thm c10-decomp, eq c10-finitesp, Prop c10-arity1) shows that these cycles with the arity-zero counterterm vertices of (Ct) tend to the tree flow of $\hbar L_1(O)^{(0)}$; the trees carrying $S^{(1)}(x)$ are attached at generic momenta and converge to the corresponding trees with edges $P(0,T)$, the pole parts cancelling as functions of ε against the same trees on the counterterm vertices. Hence (C0) and these vertices give $(\hbar L_1(O)^{(0)})[T]^{(1)}(x)$, which equals the corresponding part of the left side. (Br): Thm c8-obstruction (ii) and Prop c4-defect1 (the class I-b of DESIGN_STEP2 Part C). (Ac): the cycle and its counterterm vertices form the flow of the defect of the action in arity zero (with $S^{(1)}(x)$ on a tree) or in arity one along x; by Def c8-ob and the master equation in these arities its limit is 0, and the tree joining it to O converges.

(iii) What remains on both sides is: on the right, (C2) and the arity-one vertices of (Ct); on the left, $\big(\hbar L_1(O)^{(1)}(x)\big)[T]$ at tree level and the two counterterm trees. The vertex $-\{\hbar F_1^{(1)}(x), O\}_\varepsilon$ gives $-\{\hbar F_1^{(1)}(x), O\}$, its special-edge part tending to 0 since $F_1^{(1)}(x)$ has no poles. The tree vertex $-\{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O\}_\varepsilon$ is $\log\varepsilon$ times a local functional, by the last statement of (iv), and its special-edge part tends to 0. The vertices $-\{I, \mathrm{ct}_x^{(1)}(O)\}_\varepsilon$ and $-\{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon$ split into the brackets without special edge, which are poles and logarithms times local functionals, and their special-edge parts: with $\{f,g\}_\varepsilon - \{f,g\} = \sum_{j\geq 1} \frac{(-\varepsilon)^j}{j!}\{f,g\}_{H^j}$, the part of order ε^0 of the special-edge part of $-\varepsilon^{-m}\{f, g_m\}_\varepsilon$ is $-\frac{(-1)^m}{m!}\{f, g_m\}_{H^m}$, which is the finite special vertex (1.2); the terms with $j < m$ are poles and those with $j > m$ tend to 0; $\log\varepsilon$ times a special edge tends to 0. Each defect cycle has the expansion of Theorem 5.2: poles times local functionals, plus $[\varepsilon^0]\mathcal{D}_x$, a local functional, plus a remainder tending to 0; the trees on its outputs and the trees on the pole vertices are the same trees with edges $P(\varepsilon, T)$, so after the cancellation of the poles (iv) their difference tends to the trees with edges $P(0,T)$ of the ε^0 coefficients. The limit is therefore the tree flow of the right side of (3.1), and Thm c8-L (i) (injectivity of the tree flow) identifies the local functionals. The decomposition over the pairs with the Koszul signs is that of Thm c10-decomp (iii): a cycle passes through O along exactly two letters, the other letters being spectators.

(iv) The left side has a limit, so the local coefficients of $\log\varepsilon$ and ε^{-m} of the sum vanish. Collecting them: the poles of (C2), the poles of the special-edge parts of the (Ct) brackets shifted by j, the brackets without special edge $-s\,\mathrm{ct}_x^{(1)}(O) - \{S^{(1)}(x), \partial_\delta I^{CT}[\varepsilon](O)\} - \{I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x), O\}$, and $+\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x), O\}) + \mathrm{ct}_x^{(1)}(sO)$ moved from the left side; this gives (3.2) and (3.3). The absence of poles in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$: its coefficient of ε^{-m} is a colour-singlet local functional $\int \sum_p x^{\mu_1}\cdots x^{\mu_p} f_{\mu_1\cdots\mu_p}$ of degree -1 and total Weyl weight $4 - \frac{1}{2}\lambda_x - 2m$, the position at weight -1 and p at most the degree

of x in the position (Prop c10-bound1); a colour singlet of degree -1 contains an antifield of degree -1 and weight at least $\frac{5}{2}$ and a second adjoint field of weight at least 1, or c^* (weight 4) and c, so each density f has weight at least $\frac{7}{2}$ and the total weight is at least $\frac{7}{2}-p$. For $m\geq 1$ the total weight is at most $\frac{5}{2}$ along Σ ($p=0$), $\frac{3}{2}$ along Σ^* ($p\leq 1$), 2 along Δ ($p\leq 1$), 3 along P_b ($p=0$) and 1 along K_b ($p\leq 2$), below $\frac{7}{2}$, $\frac{5}{2}$, $\frac{5}{2}$, $\frac{7}{2}$, $\frac{3}{2}$ respectively (along Δ this is Thm c9-quadratic (i) and Thm c11-local (ii)).

(v) The argument of (i)–(iv) uses sO only through (F), which cancels between the two sides, and through the counterterm trees of the left side, which enter (iv) only. Along K_b, every step holds at fixed finite T: the edges $P(\varepsilon,T)$ are bounded uniformly in T and decay as $e^{-|y|^2/4T}$ (Def c8-PT), the graphs of (Br) and (Ac) carry at fixed ε the term $\frac{3}{8\pi^2}\hbar\log T$ times a tree with the factor $e^{-\varepsilon|p|^2}-1$ (Remark after Thm c8-K), which tends to 0 at fixed T, and Υ_{K_b} is independent of T by Theorem 5.2 (iv); Thm c8-K gives the independence of the local representative. □

Corollary 3.4 (Existence of the local representatives through arity one)**.** *If the one-loop equivariant master equation holds in arity zero and in arity one along x, then for every local functional O the local representative of $d_TO[T]$ exists in arity one along x and is given by* (3.1) *together with the arity-zero representative $L(O)^{(0)}$; along Σ, Σ^*, Δ, P_b and the rotations it is computed at $T=\infty$, along K_b at any finite T, with the same result.*

Proof. Theorem 3.3 exhibits $d_TO[T]$ in arity one along x as the tree flow of the local functional (3.1), plus the arity-one flow of $L(O)^{(0)}$; the latter exists by the same argument at arity zero. □

Remark 3.5 (Hypothesis (iii) of Def c10-hyp)**.** Corollary 3.4 removes the hypothesis that the local representative exists, used in Def c10-hyp (iii), Prop c10-arity1 and Prop c11-oddbound: along every generator it follows from the locality of the $\varepsilon\to 0$ limit of each defect cycle, which is Theorem 5.2, and from the master equation through arity one along x. The hypotheses that remain are the master equation, whose arity-one part along x is the equation $sF_1^{(1)}(x)=\mathrm{Ob}^{(1)}(x)$ for the finite counterterm, and, for $T=\infty$, the convergence of Prop c8-Tsafe.

Corollary 3.6 (Dilatation and translations)**.** *On every local functional O with $sO=0$,*

$$\sum_{a<b}\pm\Upsilon_\Delta(X_a,X_b)=\sum_{a<b}\pm\big(4\,\mathrm{lc}(X_aX_b)+C^{(2)}_\Delta(X_a,X_b)\big),\qquad \sum_{a<b}\pm\Upsilon_{P_b}(X_a,X_b)=\sum_{a<b}\pm C^{(2)}_{P_b}(X_a,X_b),$$

with the composite pair kernels $C^{(2)}_x$ of Def c11-sp (Prop c11-routes, Cor c11-D1local); the one-body terms of the two forms agree.

Proof. Both forms compute the unique local representative (Thm c8-L (i)); for $x=\Delta$ this is Prop c11-routes, whose route of the special edges is (3.1); for $x=P_b$ the same argument applies, with Cor c11-D1local and the one-body term $-\{\hbar F_1^{(1)}(P_b),-\}$ in both forms. Pair by pair the two kernels may differ by terms whose sum over the pairs of every cocycle vanishes; (3.1) holds also for $sO\neq 0$ (Theorem 3.3 (v)), while the composite form acquires the renormalised composite graphs of sO (Thm c11-local (vi)). □

Remark 3.7 (The placements along Σ, Σ^* and K_b)**.** In the classes (A)–(D) of the odd design (thm o-classes), (C2) consists of: the special edge between $S^{(1)}(x)$ and a letter (A), with $S^{(1)}(x)$ placed at the special edge; between I and $S^{(1)}(x)$ (B), the same placement at an internal vertex; between I and a letter (C), with $S^{(1)}(x)$ on a line, at no special edge, or cubic; between two vertices I (D), likewise. Along $\xi\in S_+\otimes W$ the placements are: T_1 on a line (the λ_-–c composite line of Corollary 2.3 (ii)); $T_1,\dots,T_4$ at the special edge (the antifield leg $A^*|_\perp$ of T_2 and λ^* of T_3, T_4 only there, since P has no component on them); T_5 cubic, with the leg $\lambda^*_\pm$ as an output or on the special edge. Along η the same with the Killing spinor $\rho(x)\eta$ and the term s_0 at the special edge. Along K_b: $(A|_V,A^*|_V)$ on a line (the $A|_V$–c composite line, Lemma c11-lines (iii)), and each of the five quadratic terms at the special edge. Every graph of (C2) has $k_I\geq 0$ vertices of I; $k_I=0$ occurs for the placements at the special edge only.

Remark 3.8 (Relation to the earlier design)**.** The graph classes (T), (A)–(E2) of the odd design (thm o-classes) and the endpoint analysis of k_transfer predate Chapters 8–11; checked against Def c8-renorm, Lemma c8-transfer, Def c8-rep and Thm c8-L they need the following, all built into Theorem 3.3: the vertex $\delta(Q+\{I,-\})O$ with $S^{(1)}(x)$ elsewhere (class (F), audit G3), cancelling against the arity-one part of $(sO)[T]$; the subtraction of the trees of $\hbar L_1(O)^{(0)}$ with one $S^{(1)}(x)$ (class (C0), G3); the finite parts ψ of the operators (G7), treated by Prop c8-laws (b); the statement $P_1 = M_1 = R_1 = 0$ in the covariant scheme of thm o-KD, prop o-S (G1), replaced by $P_1 \cong 0$ in the translation-covariant scheme at $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$ (Cor c11-D1fam) and by the decomposition (5.2), which keeps M_1; the canonical form with $\gamma^{\mu\nu}\partial_\mu\partial_\nu\delta$ along K_b and the exact definition of the $\varepsilon \to 0$ limit at fixed T as the T-independent coefficients of Theorem 5.2 (G5); the adjacency of two floating even insertions (G2), which enters the arity-two components only. Lemma ktr-noninv and Prop ktr-composite are used as stated; Prop o-konishi and Prop o-planevanish are Rem c11-twist and Prop c11-plane.

Remark 3.9 (Colour and Koszul signs)**.** Lemma c10-colour applies to the defect cycles with the vertex $S^{(1)}(x)$ read as follows: a quadratic term is a pairing $(-,-)$ of two adjoint fields and acts on the colour of the cycle as the identity; a cubic term T_5 has the colour form $([Z_1,Z_2],Z_3)$ and acts as ad_Y with Y its output, like a cubic vertex of I; the special edge carries $(-,-)^{-1}$ like an edge of P. Hence the colour factor of a defect cycle in the colour slots of the two letters is $(1\otimes \mathrm{ad}_{Y_r}\cdots\mathrm{ad}_{Y_1})(-,-)^{-1}$ with the outputs $Y_1,\dots,Y_r$ in the order of the chain from X_a, the output of T_5 included, and the planar reading of Lemma c10-colour (ii)–(iv) applies unchanged: a term $a\frac{\hbar}{16\pi^2}$ times this element contributes $(-1)^r\frac{a}{2}$ units to an adjacent pair of a single trace. The Koszul signs of (3.1) are those of Thm c10-decomp (iii); for odd x the component $\hbar L_1(O)^{(1)}(x)$ has the parity of O shifted by the parity of the evaluation, as in Def c8-components.

4 Census of the defect cycles and the output bound

Proposition 4.1 (Output bound along Σ, Σ^* and K_b)**.** *Let $\delta_x := -\frac{1}{2}\lambda_x$, the weight of the Killing data of x being counted as in Prop c7-select. A defect cycle of the pair (X_a,X_b) along x whose outputs have total Weyl weight w (fields only) has superficial degree*

$$D = w_{X_a} + w_{X_b} + \delta - w, \qquad \delta = \tfrac{1}{2}\ (\xi\in\Sigma,\ \textit{and the position part along } \eta), \quad \delta = -\tfrac{1}{2}\ (\Upsilon^0_\eta),$$
$$\delta = 1,0,-1\ (\textit{parts of } K_b \textit{ multiplying } v(x_O),\ dv(x_O),\ d^2v),$$

and its coefficient of ε^0 is a polynomial of degree D in the output momenta, zero for $D<0$; its coefficients of ε^{-m} have degree $D-2m$. Hence:

- *(i) along $\xi\in\Sigma$: $\Upsilon_\xi(X_a,X_b)$ is translation invariant, of Weyl weight $w_{X_a}+w_{X_b}+\frac{1}{2}$, of the pair degree and of total parity $|X_a|+|X_b|+1$, with at most $w_{X_a}+w_{X_b}+\frac{1}{2}+\deg X_a+\deg X_b$ output fields;*
- *(ii) along $\eta\in\Sigma^*$: $\Upsilon_\eta = \sum_\mu x^\mu \Upsilon_{\rho(e_\mu)\eta} + \Upsilon^0_\eta$, with Υ^0_η translation invariant of Weyl weight $w_{X_a}+w_{X_b}-\frac{1}{2}$ and at most $w_{X_a}+w_{X_b}-\frac{1}{2}$ plus the pair degree outputs;*
- *(iii) along K_b: Υ_{K_b} is a polynomial of degree at most two in the position, of total Weyl weight $w_{X_a}+w_{X_b}-1$ with the position at weight -1, of the pair degree and parity, with at most $w_{X_a}+w_{X_b}+1$ plus the pair degree outputs (the part multiplying $v(x_O)$), and the intrinsic part $\Upsilon^0_{K_b}$ has Weyl weight $w_{X_a}+w_{X_b}-1$.*

The same bounds hold for the counterterm cycles, whose outputs have the pair degree minus one, and for the finite special vertices.

Proof. Lemma c10-degree with one vertex of weight $4+\delta$: every vertex of I has weight 4, the term of $S^{(1)}(x)$ has weight $4+\delta$ on its fields and derivatives once the Killing data carry their weights, every edge of P has numerator degree w_1+w_2-2, the special edge $-|l+q|^2$ over a Schwinger interval $[0,\varepsilon]$ has degree $0=w+(4-w)-4$; the count of Lemma c10-degree gives D. The degree of the ε^0 coefficient is Theorem 5.2. Degree and parity: $S^{(1)}(x)$ has degree -1 and the special edge $+1$; $S^{(1)}(x)$ has total parity 0 for odd x and 1 for even x, the special edge 1. The number of outputs: Weyl weight plus degree is at least 1 for every field and antifield, with equality for c and A (Prop c10-bound (iii)). For the position parts, Proposition 2.6. □

Proposition 4.2 (Census). • *(i) Every defect cycle conserves the degree and shifts the total parity by that of $S^{(1)}(x)$ plus one; the centre $\mathbb{Z}/4$ of $SL(4,\mathbb{C})$, $W\mapsto iW$, is conserved once x is given its charge ($+1$ on $S_+\otimes W$ and $S_-\otimes W$, -1 on $S_-\otimes W^*$ and $S_+\otimes W^*$, 0 on the even part).*

- *(ii) Along every generator, defect cycles exist for 45 of the 55 unordered pairs of slots; the ten pairs without are the pairs of two letters among $A^*|_\perp$, λ^*_+, λ^*_-, c^*, each of which needs the special edge.*

- *(iii) $k_I=0$ (no vertex of I on the cycle) occurs exactly for the placement of $S^{(1)}(x)$ at the special edge, the cycle being letter–P–$S^{(1)}(x)$–special edge–letter; such a cycle without outputs contributes only poles.*

- *(iv) The leg sets of the defect cycles along $\xi\in S_+\otimes W$ through the pairs of field and ghost letters without derivatives are those of Table 2; the numbers of leg sets with $n\le 3$ derivatives on the two letters, along ξ and for the intrinsic parts along η and K_b, are those of Table 3. Along $S_-\otimes W^*$, $S_-\otimes W$ and for the position parts the tables are the same with the chiralities exchanged and with δ of Proposition 4.1.*

- *(v) Along $\xi\in S_+\otimes W$ the pair (c,c) carries no defect cycle at the pair weight without derivatives, and the pairs $(c,A|_V)$, $(c,A|_\perp)$, (c,λ_+), (c,λ_-) carry only leg sets with one ghost output.*

Proof. (i) Each vertex of I, each term of $S^{(1)}(x)$ and each component of P and K_0 has these charges (Table 1, Def c1-vertex, Prop c3-P). (ii)–(v) Exhaustive enumeration (`src.nosync/census_qsk.py`, on the slot data of the verified census base of Part C): chains with up to six vertices on the cycle, exactly one term of $S^{(1)}(x)$ in each of its placements and exactly one special edge, output weights up to $w_{X_a}+w_{X_b}+3+\delta$; every chain is checked for degree, parity and the centre, and for the agreement of the superficial degree with the direct count $\deg(\text{numerator})+4-2E$ (0 violations among 250172 chains along ξ, 71958 for the intrinsic part along η, 26127 for the intrinsic part along K_b, 347780 and 119553 for the even parts of degree one and zero, the output bound of Proposition 4.1 holding on every chain). A letter $A^*|_\perp$, λ^* or c^* has no component of P and must lie on the special edge, of which there is one. A cycle without outputs is a vacuum value, homogeneous in ε of degree $-D/2$ (Lemma c10-self), so it has no ε^0 term unless $D=0$. □

Table 2: the leg sets of the defect cycles along $\xi\in S_+\otimes W$ for the pairs of field and ghost letters without derivatives, with the fewest vertices of I and the number of chains in brackets; along $\xi\in S_-\otimes W^*$ exchange $\lambda_\pm$ and $\lambda^*_\pm$. A leg set contributes to Υ_ξ when its output weight is at most $W_0+\frac12$; $\emptyset$ contributes poles only.

pair	$W_0+\frac12$	defect leg sets along $\xi\in S_+\otimes W$ at the pair weight without derivatives: fewest vertices of I, number of chains

$(A\|_V, A\|_V)$	5/2	$\lambda_-\ (1;2)$, $A\|_V\,\lambda_-\ (1;8)$, $A\|_\perp\,\lambda_+\ (1;7)$, $c\,\lambda_+^*\ (1;4)$
$(A\|_V, A\|_\perp)$	5/2	$\lambda_+\ (1;4)$, $A\|_V\,\lambda_+\ (1;15)$, $A\|_\perp\,\lambda_-\ (1;15)$, $c\,\lambda_-^*\ (1;6)$
$(A\|_V, \lambda_+)$	3	$\emptyset\ (0;2)$, $A\|_V\ (0;6)$, $A\|_V^2\ (1;13)$, $A\|_\perp^2\ (1;11)$, $A\|_V^3\ (1;25)$, $A\|_V\,A\|_\perp^2\ (1;55)$, $c\,A^*\|_V\ (2;9)$, $\lambda_+\,\lambda_-\ (2;15)$
$(A\|_V, \lambda_-)$	3	$A\|_\perp\ (0;5)$, $A\|_V\,A\|_\perp\ (1;19)$, $A\|_V^2\,A\|_\perp\ (1;51)$, $A\|_\perp^3\ (1;19)$, $c\,A^*\|_\perp\ (2;2)$, $\lambda_-^2\ (2;7)$, $\lambda_+^2\ (2;6)$
$(A\|_V, c)$	3/2	$\lambda_-\,c\ (2;10)$
$(A\|_\perp, A\|_\perp)$	5/2	$\lambda_-\ (1;2)$, $A\|_V\,\lambda_-\ (1;7)$, $A\|_\perp\,\lambda_+\ (1;8)$, $c\,\lambda_+^*\ (1;2)$
$(A\|_\perp, \lambda_+)$	3	$A\|_\perp\ (0;6)$, $A\|_V\,A\|_\perp\ (1;22)$, $A\|_V^2\,A\|_\perp\ (1;59)$, $A\|_\perp^3\ (1;21)$, $c\,A^*\|_\perp\ (2;7)$, $\lambda_-^2\ (2;7)$, $\lambda_+^2\ (2;8)$
$(A\|_\perp, \lambda_-)$	3	$\emptyset\ (0;2)$, $A\|_V\ (0;5)$, $A\|_V^2\ (1;10)$, $A\|_\perp^2\ (1;11)$, $A\|_V^3\ (1;18)$, $A\|_V\,A\|_\perp^2\ (1;52)$, $\lambda_+\,\lambda_-\ (2;13)$
$(A\|_\perp, c)$	3/2	$\lambda_+\,c\ (2;9)$
(λ_+, λ_+)	7/2	$\lambda_+\ (1;3)$, $A\|_V\,\lambda_+\ (1;11)$, $A\|_\perp\,\lambda_-\ (1;9)$, $c\,\lambda_-^*\ (2;7)$, $A\|_V^2\,\lambda_+\ (2;27)$, $A\|_V\,A\|_\perp\,\lambda_-\ (2;42)$, $A\|_V\,c\,\lambda_-^*\ (2;30)$, $A\|_\perp^2\,\lambda_+\ (2;21)$, $A\|_\perp\,c\,\lambda_+^*\ (3;9)$
(λ_+, λ_-)	7/2	$\lambda_-\ (1;5)$, $A\|_V\,\lambda_-\ (1;18)$, $A\|_\perp\,\lambda_+\ (1;17)$, $c\,\lambda_+^*\ (2;7)$, $A\|_V^2\,\lambda_-\ (2;44)$, $A\|_V\,A\|_\perp\,\lambda_+\ (2;79)$, $A\|_V\,c\,\lambda_+^*\ (2;30)$, $A\|_\perp^2\,\lambda_-\ (2;40)$, $A\|_\perp\,c\,\lambda_-^*\ (2;21)$
(λ_+, c)	2	$c\ (1;5)$, $A\|_V\,c\ (1;17)$, $A\|_V^2\,c\ (2;39)$, $A\|_\perp^2\,c\ (2;18)$
(λ_-, λ_-)	7/2	$\lambda_+\ (1;2)$, $A\|_V\,\lambda_+\ (1;7)$, $A\|_\perp\,\lambda_-\ (1;8)$, $A\|_V^2\,\lambda_+\ (2;17)$, $A\|_V\,A\|_\perp\,\lambda_-\ (2;37)$, $A\|_\perp^2\,\lambda_+\ (2;19)$, $A\|_\perp\,c\,\lambda_+^*\ (2;12)$
(λ_-, c)	2	$A\|_\perp\,c\ (1;10)$, $A\|_V\,A\|_\perp\,c\ (2;44)$
(c, c)	1/2	none

Table 3: the number of leg sets of the defect cycles contributing to Υ_ξ ($\xi \in S_+ \otimes W$), to the intrinsic part Υ^0_η ($\eta \in S_+ \otimes W^*$) and to the intrinsic part $\Upsilon^0_{K_b}$, for n derivatives on the two letters, up to six vertices on the cycle. The ten pairs of two letters among $A^*|_\perp$, $\lambda_\pm^*$, c^* carry no defect cycle.

pair	along ξ, $n = 0..3$				Υ^0_η, $n = 0..3$				$\Upsilon^0_{K_b}$, $n = 0..3$			
$(A\|_V, A\|_V)$	4	9	20	43	1	4	9	20	2	4	8	19
$(A\|_V, A\|_\perp)$	4	9	20	43	1	4	9	20	1	2	7	16
$(A\|_V, \lambda_+)$	8	19	35	68	2	7	16	33	1	4	9	20
$(A\|_V, \lambda_-)$	7	16	33	61	4	8	19	35	1	4	9	20
$(A\|_V, c)$	1	4	9	20	0	1	4	9	1	2	4	8
$(A\|_V, A^*\|_V)$	8	22	49	97	3	8	22	49	1	6	15	35
$(A\|_V, A^*\|_\perp)$	8	22	49	97	3	8	22	49	1	5	13	29
$(A\|_V, \lambda_+^*)$	3	11	26	62	1	6	15	35	1	3	8	21
$(A\|_V, \lambda_-^*)$	6	15	35	73	0	3	11	26	1	3	8	21
$(A\|_V, c^*)$	3	13	34	80	0	3	13	34	1	3	10	25
$(A\|_\perp, A\|_\perp)$	4	9	19	39	1	4	9	19	2	4	7	14
$(A\|_\perp, \lambda_+)$	7	16	33	61	4	7	16	30	1	3	7	16
$(A\|_\perp, \lambda_-)$	7	16	30	60	2	7	16	33	1	3	7	16
$(A\|_\perp, c)$	1	4	9	20	0	1	4	9	0	1	2	7
$(A\|_\perp, A^*\|_V)$	8	22	49	97	3	8	22	49	1	5	13	29
$(A\|_\perp, A^*\|_\perp)$	7	18	40	81	3	7	18	40	0	1	5	16
$(A\|_\perp, \lambda_+^*)$	3	10	26	59	1	5	14	31	0	1	4	14

$(A\|_\perp, \lambda^*_-)$	5	14	31	70	0	3	10	26	0	1	4	14
$(A\|_\perp, c^*)$	3	13	34	80	0	3	13	34	0	1	4	19
(λ_+, λ_+)	9	20	42	79	3	7	16	35	2	6	12	24
(λ_+, λ_-)	9	20	43	82	4	9	20	43	4	7	15	27
(λ_+, c)	4	8	19	35	1	2	7	16	0	1	4	9
$(\lambda_+, A^*\|_V)$	15	35	73	143	5	14	31	70	3	8	21	46
$(\lambda_+, A^*\|_\perp)$	14	31	70	135	3	10	26	59	1	4	14	35
(λ_+, λ^*_+)	8	21	47	88	3	8	21	47	0	2	7	19
(λ_+, λ^*_-)	8	21	46	86	1	3	11	29	0	1	5	14
(λ_+, c^*)	10	25	62	116	2	5	19	49	0	2	10	29
(λ_-, λ_-)	7	16	35	68	4	9	20	42	2	6	12	24
(λ_-, c)	2	7	16	33	2	4	8	19	0	1	4	9
$(\lambda_-, A^*\|_V)$	14	31	70	135	6	15	35	73	3	8	21	46
$(\lambda_-, A^*\|_\perp)$	10	26	59	123	5	14	31	70	1	4	14	35
(λ_-, λ^*_+)	3	11	29	60	3	8	21	46	0	1	5	14
(λ_-, λ^*_-)	8	21	47	88	3	8	21	47	0	2	7	19
(λ_-, c^*)	5	19	49	101	3	10	25	62	0	2	10	29
(c, c)	0	1	4	9	0	0	1	4	0	1	2	4
$(c, A^*\|_V)$	9	20	43	82	4	9	20	43	4	8	19	35
$(c, A^*\|_\perp)$	9	20	43	82	4	9	20	43	2	7	16	32
(c, λ^*_+)	6	14	30	53	4	8	19	35	1	4	9	20
(c, λ^*_-)	8	19	35	64	2	6	14	30	1	4	9	20
(c, c^*)	8	22	49	91	3	8	22	49	1	6	15	35
$(A^*\|_V, A^*\|_V)$	13	34	80	169	3	13	34	80	3	10	25	61
$(A^*\|_V, A^*\|_\perp)$	13	34	80	169	3	13	34	80	1	4	19	49
$(A^*\|_V, \lambda^*_+)$	5	19	49	101	3	10	25	62	0	2	10	29
$(A^*\|_V, \lambda^*_-)$	10	25	62	116	2	5	19	49	0	2	10	29
$(A^*\|_V, c^*)$	4	15	46	101	1	4	15	46	0	1	7	24

Remark 4.3 (The pairs of the holomorphic twist)**.** With the letter dictionary of Rem c11-twist ($\gamma \leftrightarrow A_{ij}z^j$, $\beta \leftrightarrow u'^\alpha \lambda_\alpha{}^i$ modulo z^i, $\partial_{\dot\alpha} c \leftrightarrow \bar\lambda_{\dot\alpha i} z^i$, $c \leftrightarrow c$, $b \leftrightarrow u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta$ or a component of λ^*_+, $\partial_{\dot\alpha} \leftrightarrow u'^\alpha \sigma^m_{\alpha\dot\alpha} D_m$), the pair entries of [BK, (3.64)–(3.69)] are carried by the following pairs and leg sets along $\xi = u \otimes z \in S_+ \otimes W$ (Table 4); every listed leg set is present in the census, at the output weight $w_{X_a} + w_{X_b} + \frac{1}{2}$ required by Lemma arity of the targets.

Table 4: the defect cycles carrying the pair entries of the holomorphic twist. The entry bc with b represented by the self-dual field strength has arity two (Table c11-twist) and is carried by the cycles with two vertices $S^{(1)}(\xi)$ of Section 7.

entry	pair of slots	n	leg set (output weight $W_0 + n + \frac{1}{2}$)	chains, fewest k_I
$\beta\gamma$ [BK, (3.66)]	$(\lambda_+, A\|_\perp)$	0	$\lambda_-\lambda_-$	7, 2
$\beta\beta$ [BK, (3.67)]	(λ_+, λ_+)	0	$A\|_\perp\lambda_-$	9, 1
$b\gamma$ [BK, (3.68)]	$(A\|_V, A\|_\perp)$	1	$A\|_\perp\lambda_-$	15, 1
βb [BK, (3.69)]	$(\lambda_+, A\|_V)$	1	$\lambda_+\lambda_-$; $A\|_\perp A\|_\perp$	15, 2; 11, 1
bb [BK, (3.65)]	$(A\|_V, A\|_V)$	2	$A\|_V\lambda_-$; $A\|_\perp\lambda_+$	8, 1; 7, 1

bc [BK, (3.64)], $b \leftrightarrow \lambda_+^*$	(c, λ_+^*)	0	$\lambda_-\lambda_-$	1, 2
$\gamma\gamma$, γc, βc, cc [BK, (3.11)]	$(A\|_\perp, A\|_\perp)$; $(A\|_\perp, c)$; (λ_+, c); (c, c)	0	$A\|_\perp\lambda_+$, $A\|_V\lambda_-$, $c\lambda_+^*$, λ_-; $c\lambda_+$; c, $A\|_V c$; none	

5 Closed forms of the two-body kernels

Definition 5.1 (Cycle data of a defect cycle)**.** For a defect cycle with n vertices (k_I of I and V_s) and edges $e_0, \dots, e_n$, e_0 leaving X_a, let q_j be the sum of the incoming momenta of the outputs of the first j vertices, so that e_j carries $l + q_j$ (Def c10-cycle). Along η and K_b a momentum r is injected at V_s and carried by $e_s, \dots, e_n$ to X_b and to ω. The Schwinger times are $t_\sigma \in [0, \varepsilon]$ on the special edge and $t_j \in [\varepsilon, T]$ on the other edges; $\tau = \sum_j t_j$, $\alpha_j = t_j/\tau$ on the simplex of dimension n. The numerator is the product of the vertex functions of I (Def c1-vertex), the function of the term of $S^{(1)}(x)$ at V_s (Table 1, Def c7-vdot), the edge factors of Rem c10-numerator, and the special-edge factor $-|l + q_\sigma|^2$ times the fibre factor of K_0 (Lemma c10-special, Rem c11-twocomp); the centre, the momentum form and the m-th simplex integrand F_m are those of Def c10-cycle with n in place of k.

Theorem 5.2 (Defect cycles as simplex integrals)**.**

- *(i) At generic output momenta and $T = \infty$, for $x \in \Sigma \cup \Sigma^* \cup \{\Delta, P_b, M_\omega\}$, the value of a defect cycle is $\sum_{m\geq 1} \varepsilon^{-m} a_m + a_0 + o(1)$ with*

$$
\begin{aligned}
16\pi^2 a_m &= \frac{1}{m} \int_{\alpha_\sigma \ \text{smallest}} \Big[\big(\min_{f\neq\sigma} \alpha_f \big)^m - \alpha_\sigma^m \Big] F_m \, d\alpha \quad (m \geq 1), \\
16\pi^2 a_0 &= \int_{\alpha_\sigma \ \text{smallest}} F_0 \, \log \frac{\min_{f\neq\sigma} \alpha_f}{\alpha_\sigma} \, d\alpha,
\end{aligned}
\tag{5.1}
$$

  and no term $\log\varepsilon$. Each a_m is a polynomial in the output momenta (and in r) of degree $D - 2m$, D the superficial degree of Proposition 4.1; $a_0 = 0$ for $D < 0$.

- *(ii) Along η, the position part along b is the same formula with $S^{(1)}(\rho(b)\eta)$ at V_s; the intrinsic part is $-i\sum_\mu \partial_{r_\mu}$ at $r = 0$ of the formula with $S^{(1)}(\xi_\mu)$, $\xi_\mu = \rho(e_\mu)\eta$, plus the formula with s_0 at V_s.*

- *(iii) Along K_b, at every finite T, the coefficients a_m, a_0 of the $\varepsilon \to 0$ expansion are given by (5.1); the parts of Proposition 2.6 (ii) are obtained from the formula along $P_{v(x_O)}$, from the translation-invariant parts along Δ and $M_{\omega_{x_O}}$, and from the intrinsic vertex $S^{(1)}(K_{b,x_O})$, whose monomials $z^\mu z^\nu$ and z^μ act as $-\partial_{r_\mu}\partial_{r_\nu}$ and $-i\partial_{r_\mu}$ at $r = 0$.*

- *(iv) $T\partial_T$ of every defect cycle is $O(\varepsilon)$ uniformly for T in compact subsets of $(0, \infty)$; hence Υ_{K_b} is independent of T, and so are the coefficients a_m.*

Proof. (i) The domain $\{t_\sigma \leq \varepsilon,\ t_f \geq \varepsilon\ (f \neq \sigma)\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq\sigma}\alpha_f$ and $1/\min_f \alpha_f$ (Lemma c10-special), which differ on the region where α_σ is the smallest coordinate. Prop c10-shape gives, for each shape, the poles, the coefficient of $\log\varepsilon$ (the same for both shapes) and the finite part $-\int F_0 \log(\text{shape})$ plus a shape-independent remainder; the difference is (5.1). The region $\tau \geq 1$ converges at $T = \infty$: the edges carry distinct generic momenta (proof of Thm c10-decomp (ii)), the composite line along x is of order $|p|^{-2}$ at small momentum along Σ (momentum degree $\frac{3}{2} + 0 - 4 + \frac{1}{2}$, Corollary 2.3 (iv)), of order $|p|^{-3}$ along Δ, P_b, M_ω (Prop c9-composite (iv)), of order $|p|^{-2}$ for the position part along η (a constant spinor $\rho(b)\eta$) and of order $|p|^{-3}$ for its intrinsic part (one derivative in r), and the measure of $\{\alpha_\sigma \leq \varepsilon/\tau\}$ is of order ε; the remainder is $O(\varepsilon)$ times convergent integrals, as in Thm c11-simplex. The degree: the coefficient of τ^{-1-m} after the Gaussian integral is homogeneous of degree $D - 2m$. (ii) Proposition 2.6 (i); the derivative in r commutes

with the Gaussian integral and the simplex integration, r entering the centre and the momentum form polynomially. (iii) Proposition 2.6 (ii); at finite T the domain is $t_j \in [\varepsilon, T]$, and the region $\tau \geq \tau_0$ is compact in τ, so the $\varepsilon \to 0$ expansion is that of the small-τ region, independent of T, plus $O(\varepsilon)$. (iv) $T\partial_T$ replaces one edge e by TL_T, of scalar factor $Te^{-T|l+q_e|^2}$, which confines the loop momentum; the special edge carries $e^{-\varepsilon|l+q_\sigma|^2} - 1 = O(\varepsilon(1+|l|)^2)$, and the polynomial position dependence of the Killing data enters as derivatives in r, which preserve the bounds (k_transfer Thm ktr-defect (ii)). The finite special vertices are local functionals of the coefficients of ct_m, $\mathrm{ct}^{(1)}_{x,m}$, which are independent of T (Prop c8-Tindep (i)). □

Proposition 5.3 (Generating functions in the point splitting)**.** *With the letters at $x + y_a$, $x + y_b$, $y = y_b - y_a$, the m-th simplex integrand of a defect cycle with n vertices for the numerator (chain numerator) $e^{i(y,l)}$ is the chain numerator with l replaced by $-i\partial_y$ applied to*

$$(-1)^{m+n-1} e^{i(y,centre)} \Big(\frac{|y|}{2\sqrt{momentum\ form}} \Big)^{m+n-1} I_{m+n-1}\big(|y|\sqrt{momentum\ form}\big)$$
$$= (-1)^{m+n-1} e^{i(y,centre)} \sum_{i\geq 0} \frac{(|y|^2/4)^{i+m+n-1} (momentum\ form)^i}{i!\,(i+m+n-1)!},$$

and

$$\sum_{\alpha,\beta} \frac{y_a^\alpha y_b^\beta}{\alpha!\,\beta!}\, \Upsilon_x\big(\partial^\alpha \varphi_a, \partial^\beta \varphi_b\big) = \frac{\hbar}{16\pi^2}\, e^{i(y_b,q_n)} \sum_{chains} \int_{\alpha_\sigma\ \ smallest} F_0 \log \frac{\min_{f\neq\sigma} \alpha_f}{\alpha_\sigma}\, d\alpha$$
$$+ \sum_{\alpha,\beta} \frac{y_a^\alpha y_b^\beta}{\alpha!\,\beta!}\, \mathcal{F}_x\big(\partial^\alpha \varphi_a, \partial^\beta \varphi_b\big),$$

the integrands taken for the generating numerator; the coefficients of ε^{-m} have the same generating function with the weights of (5.1)*. In position space iq_j is the derivative of the product of the outputs at x.*

Proof. Prop c10-genfun with $n+1$ edges: the Gaussian integral gives τ^{-2}, the Schwinger measure $\tau^n d\tau\, d\alpha$, so the power τ^{-1-m} comes from the term $j = i+m+n-1$ of the Laplacian series; Theorem 5.2 replaces the weight $(\min_e \alpha_e)^m$ by the weights of the region where α_σ is smallest. The letter derivatives are generated by $e^{-i(y_a,l)+i(y_b,l+q_n)}$. □

Proposition 5.4 (One and two vertices in closed form)**.**

- *(i) $n = 1$ ($k_I = 0$, V_s at the special edge): with $\alpha := \alpha_\sigma \in [0, \frac{1}{2}]$, the centre and the momentum form are those of the bubble, $-\alpha q$ and $\alpha(1-\alpha)|q|^2$ for the total output momentum q (nonzero only for a cubic term T_5 with its output on the cycle side), and*

$$16\pi^2 a_0 = \int_0^{1/2} F_0(\alpha) \log \frac{1-\alpha}{\alpha}\, d\alpha, \qquad 16\pi^2 a_m = \frac{1}{m} \int_0^{1/2} \big((1-\alpha)^m - \alpha^m\big) F_m(\alpha)\, d\alpha,$$

rational combinations of 1 and $\log 2$ for polynomial numerators. For the numerator $-|l+q|^2$ on the special edge and 1 on the other edge, $16\pi^2$ times the value is $-\frac{1}{2\varepsilon} - \frac{3}{8}|q|^2 + o(1)$ (Lemma c10-special).

- *(ii) $n = 2$ with V_s on a line (the composite line along x between X_a or a vertex of I and the special edge): Prop c11-k1 applies verbatim, with $\beta = \alpha_\sigma \in [0, \frac{1}{3}]$, the weights $W_0(\beta) = (1-\beta)\log\frac{1-\beta}{2\beta} - (1-3\beta)$ and $W_m(\beta)$ of Prop c11-k1, and values in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$, the two segments of the line carrying the same momentum at $r = 0$.*

- *(iii) In general the coefficients are integrals of polynomials in α times* log *of ratios of coordinates over the rational subsimplices $\{\alpha_\sigma \le \alpha_f \le \cdots\}$ of the region where α_σ is smallest, evaluated exactly by ordering the coordinates and integrating Dirichlet monomials against* log *(the method of the Part-C engines).*

Proof. (i) The simplex is $[0,1]$, the region where α_σ is smallest is $\alpha_\sigma \le \frac{1}{2}$, and $\min_{f\neq\sigma}\alpha_f = 1-\alpha_\sigma$; $\int_0^{1/2}\alpha^a(1-\alpha)^b\log\frac{1-\alpha}{\alpha}\,d\alpha \in \mathbb{Q}+\mathbb{Q}\log 2$. The example is the computation of Lemma c10-special with the roles of the edges relabelled. (ii) Prop c11-k1, whose proof uses only that the two segments adjacent to the internal vertex carry the same momentum and that the special edge is the third edge. (iii) Thm c11-simplex and Prop c11-k1. □

Example 5.5 (The cubic term at the special edge)**.** The only defect cycles with $k_I = 0$ and outputs are those with a cubic term of $S^{(1)}(\xi)$ at the special edge: for $\xi \in S_+ \otimes W$ and the pair $(A|_\perp, \lambda_+)$, the cycle $A|_\perp \xrightarrow{P} T_5(A|_\perp, A|_\perp, \lambda_+^*) \xrightarrow{K_\varepsilon - K_0} \lambda_+$ with the output $A|_\perp$ of momentum q. By Proposition 5.4 (i) its value is $\frac{\hbar}{16\pi^2}\big(-\frac{1}{2\varepsilon}-\frac{3}{8}|q|^2\big)$ times the value of T_5 on the fibre elements and colour factors of the two letters and of the output, with the fibre factor $2(-,-)^{-1}$ of the $A|_\perp \otimes A|_\perp$ edge and the inverse fibre pairing of (λ_+,λ_+^*) on the special edge; its contribution to $\Upsilon_\xi(A|_\perp,\lambda_+)$ on the leg set $A|_\perp$ is $\frac{3}{8}\frac{\hbar}{16\pi^2}\,\partial^2$ of the output, times that factor, $|q|^2$ being $-\partial^2$. Its pole $-\frac{1}{2\varepsilon}$ enters (3.3) against ct_1 of the pair (λ_+,λ_+) of the classical transformation $\{S^{(1)}(\xi), A|_\perp\} \ni +\Gamma(\xi,\lambda)|_\perp$, whose coefficient of ε^{-1} has the leg set $A|_\perp$ (Table c10-legsets).

Proposition 5.6 (The special conformal component at a finite endpoint)**.**

- *(i)* $K_1(O) = \sum_{a<b}\pm\Upsilon_{K_b}(X_a,X_b) - \{\hbar F_1^{(1)}(K_b), O\}$ *is the same local functional at every $T \in (0,\infty)$ and in the limit $T\to\infty$ (Thm c8-K), and, at the point x of the density,*

$$K_1(O) = v(x)^\mu\, P_1(O)\big|_{P_{e_\mu}} - (b,x)\, D_1^0(O) + M_1^0(O)\big|_{\omega_x} + K_1^0(O), \tag{5.2}$$

with D_1^0, M_1^0 the translation-invariant parts of D_1 and of the rotation component along ω_x, and $K_1^0(O)$ the intrinsic part, a translation-invariant local functional of Weyl weight $w_O - 1$, in a scheme in which the finite counterterm $\hbar F_1^{(1)}(K_b)$ decomposes as $S^{(1)}(K_b)$ in (2.7). On plane waves the canonical form of Def c11-canonical is extended by $\gamma^{\mu\nu}\partial_\mu\partial_\nu\delta$, and $\int\omega\, x^\mu x^\nu K_{\mu\nu}$ corresponds to $-\gamma^{\mu\nu}$.

- *(ii) The observables at scale T keep, in their component along K_b, the term $\frac{3}{8\pi^2}\hbar\log T$ times $\sum_a \partial_{A|_V = b\otimes X_a}\partial_{c=X^a}$ of the order-$\hbar^0$ part of $O[\infty]$ (Thm c8-K, Prop c8-Tsafe); this term is not part of K_1. In the defect graphs at fixed ε the coefficient of $\log T$ is $\frac{3}{8\pi^2}\hbar$ times Q of the zero-mode tree of $\partial_\delta W(P(\varepsilon,\infty), I+\delta O)$, of order ε (Remark after Thm c8-K); it vanishes as $\varepsilon\to 0$ at fixed T.*

- *(iii) In the translation-covariant scheme at $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, on the families of Def c11-families, $P_1 \cong 0$ (Cor c11-D1fam); if moreover $M_1 \cong 0$ on the family,* (5.2) *gives on cohomology*

$$K_1(O) \cong -(b,x)\, D_1(O) + K_1^0(O),$$

the conformal expectation of Rem c7-KDelta ($-(b,x) = \frac{1}{8}\operatorname{div} K_b$) plus the intrinsic part, which vanishes on Lorentz-scalar families (the scalar bilinears and the letters, Cor c7-K) and is in general fixed by the relations of (iv).

- *(iv) Three relations fix K_1^0 on cohomology: the arity-two component of $L\circ L = 0$ at (η_1,η_2) with $\Gamma(\eta_1,\eta_2)|_V = b$, which expresses K_1 along K_b through S_1 and the classical transformations along η_1, η_2 (Remark after Thm c8-K); the relation $K = P^\dagger$ of Cor c8-gram, which in a scheme with $P_1 \cong 0$ reads K_1 along $K_{\bar b} = \hbar\big[\{S^{(1)}(K_{\bar b}),-\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big]$ (Cor c11-K); and the component at (K_b,Δ), which states that K_1 lowers the Weyl weight by one.*

Proof. (i) Theorem 3.3 (v), Theorem 5.2 (iv), Proposition 2.6 (ii), (iii); the one-body terms decompose with $\hbar F_1^{(1)}(K_b)$ in the stated scheme. The canonical form: $\int \omega\, x^\mu x^\nu e^{i(k,x)}$ is $-\partial_{k_\mu}\partial_{k_\nu}$ of $\hat{\omega}$. (ii) Thm c8-K and the Remark after it. (iii) Cor c11-D1fam and Prop c11-freedom (v) give $P_1 \cong 0$; D_1^0 and the position part of D_1 combine with $v(x)^\mu P_1$ to $-(b,x)D_1$ up to $x^2(b,P_1)$, by Prop c11-position (i) (the position part of D_1 is twice P_1); Cor c7-K excludes the second-derivative term on scalar bilinears and letters. (iv) Cor c8-homotopy (ii) at (η_1,η_2) with $[\eta_1,\eta_2] = -K_{\Gamma(\eta_1,\eta_2)|_V}$ (Prop c2-special), Cor c8-gram, and Cor c8-homotopy (ii) at (K_b,Δ) with $[K_b,\Delta] = -2K_b$. □

Remark 5.7 (The K-route and $K = P^\dagger$)**.** The K-route of Proposition 5.6 (i) computes K_1 from the defect cycles at a finite endpoint; $K = P^\dagger$ computes it from P_1 and the one-loop inner product. The two agree when the scheme is reflection compatible and the Ward identity of Thm c8-adjoint holds. On a scalar primary at the origin both sides vanish: $K_1^0 = 0$ by Cor c7-K, and the commutator $[\{S^{(1)}(K_{\bar{b}}),-\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1]$ annihilates a primary at the origin on which $\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1$ is diagonal. On its descendants $\partial_\mu O$ the commutator is the difference of the two-point normalisations at weights w_O+1 and w_O times the classical transformation, and the K-route gives the intrinsic kernel on the descendant pairs: this is the cross-check, available once Gram_1 is computed from the two-point graphs of Def c8-2pt.

6 Renormalisation freedom and invariant content

Definition 6.1 (The arity-one finite counterterm along Σ)**.** For $\xi \in \Sigma$ write the quadratic part of $S^{(1)}(\xi)$ as $S_2^V(\xi)+S_2^\perp(\xi)$, with $S_2^V(\xi) = T_1+T_3$ the part with the legs $(\lambda, A^*|_V)$, $(A|_V,\lambda^*)$ and $S_2^\perp(\xi) = T_2+T_4$ the part with the legs $(\lambda, A^*|_\perp)$, $(A|_\perp,\lambda^*)$ (Table 1). The arity-one equation along ξ is

$$(Q+\{I,-\})\,\hbar F_1^{(1)}(\xi) + \big\{S^{(1)}(\xi), \hbar F_1^{(0)}\big\} = \hbar\,\mathrm{Ob}^{(1)}(\xi), \tag{6.1}$$

the arity-one part along ξ of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ (Thm c8-obstruction (ii)); $\hbar\,\mathrm{Ob}^{(1)}(\xi)$ is the local functional whose tree flow is the order-$\hbar$ limit of the arity-one defect of the action along ξ: the cycles of vertices of I with one vertex $S^{(1)}(\xi)$ and one special edge (the classes (B), (D) without O), and the counterterm vertices $-QI^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\xi) - \{I, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\xi)\}_\varepsilon - \{S^{(1)}(\xi), I^{CT}[\varepsilon]\}_\varepsilon$ (Prop o-uv).

Proposition 6.2 (The quadratic free directions along Σ)**.**

- *(i) The quadratic,* $\mathrm{Spin}(4)\times\mathrm{Spin}(6)\times\mathfrak{g}$*-covariant local functionals of degree* -1 *and Weyl weight* $\frac{9}{2}$ *(fields and derivatives), linear in* $\xi \in S_+\otimes W$ *and translation invariant, are spanned by the five functionals* T_1, T_2, T_3, T_4 *and* $\int\big((d^*A|_V)\,\xi,\lambda^*\big)$*; their* Q*-closed combinations are spanned by* $S_2^V(\xi)$ *and* $S_2^\perp(\xi)$*, and no nonzero combination is* Q*-exact.*

- *(ii) At fixed* $\hbar F_1^{(0)}$*, two solutions of* (6.1) *differ by a* $(Q+\{I,-\})$*-closed local functional of degree* -1 *linear in* ξ*; in the quadratic sector by* $\theta_V^\xi S_2^V(\xi) + \theta_\perp^\xi S_2^\perp(\xi)$*. The cubic part of* (6.1) *then contains* $\theta_V^\xi\{I_3, S_2^V(\xi)\} + \theta_\perp^\xi\{I_3, S_2^\perp(\xi)\}$*. For* $\theta_V^\xi = \theta_\perp^\xi =: \theta^\xi$ *the direction extends, with the cubic term* $\theta^\xi S_3(\xi)$*, to the closed direction* $\theta^\xi S^{(1)}(\xi)$*, since* $\{I, S^{(1)}(\xi)\} = -QS^{(1)}(\xi)$*: the arity-one value of* G_B *(Def c8-B) and of the field redefinition by the R-rotation* $S^{(1)}(R_\xi)$*,* $[R_\xi,\xi] = \xi$ *(Thm c8-twist). The difference* $\theta_V^\xi - \theta_\perp^\xi$ *is fixed by the cubic part of* (6.1) *when the class of* $\{I_3, S_2^V(\xi)\}$ *modulo* Q *of the cubic local functionals of degree* -1 *is nonzero, as for the sector coefficients along* Δ *(Prop c11-arity1 (iii)). Field redefinitions by the field countings change* $\hbar F_1^{(0)}$ *and* $\hbar F_1^{(1)}(\xi)$ *together, the latter by* $\{S^{(1)}(\xi), N_y\}$*, which rescales each arrow (Proposition 6.3 (a)).*

- *(iii) The one-body kernel of a letter* X *along* ξ *is* $-\{\hbar F_1^{(1)}(\xi), X\}$*; the free directions add* $-\theta_V^\xi\{S_2^V(\xi), X\} - \theta_\perp^\xi\{S_2^\perp(\xi), X\}$*, which on the letters* $A|_V$*,* λ *(the* V*-arrow),* $A|_\perp$*,* λ *(the*

$\perp$-arrow) is $-\theta_V^\xi$ or $-\theta_\perp^\xi$ times the corresponding arrow of the classical transformation (Lemma 2.1: $\{S_2^V(\xi), A|_V\} = +\Gamma(\xi,\lambda)|_V$, $\{S_2^\perp(\xi), A|_\perp\} = +\Gamma(\xi,\lambda)|_\perp$).

The same holds along $S_- \otimes W^*$ with the chiralities exchanged and along Σ^* for the position part with $\rho(b)\eta$ in place of ξ. Along Σ^*, with the position included, the Q-closed quadratic directions are $\sum_\mu \int x^\mu S_2^V(\xi_\mu)$ and $\sum_\mu \int x^\mu S_2^\perp(\xi_\mu) + \int s_0$: the intrinsic term s_0 is not an independent closed direction (route B, task TQ0 along Σ^*); the sum of the two is the quadratic part of $S^{(1)}(\eta)$.

Proof. (i) Field contents of degree -1 and field weight $\frac{9}{2}$ linear in a constant spinor: (λ, A^*) without derivatives and (A, λ^*) with one derivative (c and c^* would need weight $\frac{1}{2}$); covariant pairings: $\Gamma(\xi,\lambda)$ with A^* in its V and $W_\mathbb{R}^\perp$ parts, and $\rho(\partial A)\xi$ with λ^*, where ∂A is $dA|_V$, $d^*A|_V$ or the mixed part $\rho(e_m)\rho(\partial_m A|_\perp)$; a factor $\rho(e_1e_2e_3e_4)$ acts on each chirality summand by a sign and gives no new structure. On functionals Q substitutes $A \mapsto -dc$, $A^* \mapsto -\frac{1}{2}d^*dA$, $\lambda^* \mapsto \not{d}\lambda$, $c^* \mapsto d^*(A^*|_V)$ and annihilates c, λ (Prop c1-Q); Q of $\int((d^*A|_V)\xi, \lambda^*)$ contains $-\int((d^*dc)\xi, \lambda^*)$, produced by no other structure, so its coefficient vanishes in a closed combination; the remaining four map to (λ, A)-functionals, and Q preserves the split into the parts along V and along $W_\mathbb{R}^\perp$, so the closed space is at most two-dimensional; $S_2^V(\xi)$ and $S_2^\perp(\xi)$ are closed, being the two parts of the quadratic part of $\{S, S^{(1)}(\xi)\} = 0$. A Q-exact functional would be Q of a functional of degree -2 and field weight $\frac{9}{2}$, which does not exist (two antifields have weight at least 5, and c^* needs a further field of weight $\frac{1}{2}$). (ii) Thm c8-obstruction (iii) restricted to arity one along ξ with the arity-zero part fixed; the cubic part of $s(\theta_V^\xi S_2^V + \theta_\perp^\xi S_2^\perp)$ is the bracket with I_3, and $\{I_3, S_2(\xi)\} = -QS_3(\xi)$ for the full quadratic part $S_2 = S_2^V + S_2^\perp$ (the cubic part of $\{S, S^{(1)}(\xi)\} = 0$); $G_B^{(1)}(\xi) = S^{(1)}(B\xi) = S^{(1)}(\xi)$ for $\xi \in S_+ \otimes W$ (Def c8-B); the last statement is Prop c8-laws (a) with $\phi^{(0)} = N_y$. (iii) Theorem 3.3 (iii) and Lemma 2.1. The engine task TC0-Σ of Section 9 confirms (i) by linear algebra on plane-wave normal forms, as Prop c11-tc0 does along Δ. □

Proposition 6.3 (Transformation laws of Q_1, S_1, K_1)**.** *Under the changes of Prop c8-laws the components change as follows; $\{S^{(1)}(x), -\}$ is the classical transformation.*

- *(a) Field redefinition $F \mapsto F + \hbar s\phi$: the component along x changes by $-\{(s\phi)^{(1)}(x), -\}$, $(s\phi)^{(1)}(x) = \pm\{S^{(1)}(x), \phi^{(0)}\} + (Q + \{I, -\})\phi^{(1)}(x)$: a one-body change. For the field countings $\phi^{(0)} = \sum_f y_f \int(f, f^*)$ the change of Q_1 along ξ multiplies each arrow $f \to f'$ of the classical transformation by $y_{f'} - y_f$ (Prop c8-B proof); for $\phi^{(0)} = t\,S^{(1)}(\Delta)$, $(s\phi)^{(1)}(x) = t\{S^{(1)}(x), S^{(1)}(\Delta)\}$ is $\pm t\lambda_x S^{(1)}(x)$, the sign fixed by $\{S^{(1)}(\xi), S^{(1)}(\Delta)\} = S^{(1)}(\xi)$ and $\{S^{(1)}(P_b), S^{(1)}(\Delta)\} = -2S^{(1)}(P_b)$ (proof of Prop c8-B), so the change is a multiple of the classical transformation proportional to λ_x, zero along Δ and the rotations.*

- *(b) Finite parts of the operators $\psi \mapsto \psi + \psi'$: the component along x changes by $[\{S^{(1)}(x), -\}, \psi']$, i.e. on O by $\{S^{(1)}(x), \psi'(O)\} - \psi'(\{S^{(1)}(x), O\})$. For $\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$ this is $\frac{a}{2}\lambda_x\{S^{(1)}(x), -\}$: $-\frac{a}{2}$ along Σ, $+\frac{a}{2}$ along Σ^*, $+a$ along K_b (Prop c9-scheme), a one-body change. A finite part ψ' of the operators of the Weyl weight of $\{S^{(1)}(x), O\}$ that vanishes on O changes the component by $-\psi'(\{S^{(1)}(x), O\})$, a change of the two-body kernels: for the Konishi descendant $O_\zeta = \int \omega\,(u'^\alpha \lambda_\alpha{}^i, A_{ij}z^j)$, whose classical component along $\xi = u \otimes z$ is κ_B times the cubic $B_z = \int \omega\,([A^{ik}, A_{kj}z^j], A_{il}z^l)$, the finite part $\psi'(B_z) = y\,\bar{F}_z$, $\bar{F}_z = \int \omega\,(\bar{\lambda}_{\dot{\alpha}i}z^i, \bar{\lambda}^{\dot{\alpha}}{}_j z^j)$, changes the Konishi entry c_K by $-\kappa_B y$ (Rem c11-twist, Prop o-konishi).*

- *(c) Closed directions: the coupling direction leaves Q_1, S_1, K_1 unchanged and changes $L_1^{(0)}$ by $-a(Q + \{I, -\})$ and $L_1^{(2)}$ by $a\{S^{(2)}, -\}$; the outer direction $F \mapsto F + \hbar a G_B$ changes Q_1 along $S_+ \otimes W$ by $-a\{S^{(1)}(\xi), -\}$ and along $S_- \otimes W^*$ by $+a\{S^{(1)}(\xi), -\}$, S_1 along $S_+ \otimes W^*$ by $+a\{S^{(1)}(\eta), -\}$ and along $S_- \otimes W$ by $-a\{S^{(1)}(\eta), -\}$, and leaves K_1, D_1, P_1, M_1, R_1 unchanged (Prop c8-laws (c)).*

- *(d) The free directions of $\hbar F_1^{(1)}(x)$: along Σ the one-body changes of Proposition 6.2 (iii); along Δ, P_b the sector coefficients $\theta, \theta_\lambda, \theta_V, \theta_P, \dots$ of Def c11-free, which enter D_1, P_1, and through P_1 the relations of S_1 to Q_1 (Prop c11-S (ii)) and of K_1 to D_1 (Proposition 5.6 (iii)); along K_b, s-closed directions of degree -1 quadratic in the Killing data, which change K_1 by one-body terms.*

- *(e) The cutoff shape (Prop c10-shape) changes the finite parts of the defect cycles, of the special-edge cycles of $L_1^{(0)}$ and the finite special vertices; by the uniqueness of the counterterms up to finite local terms [C07, Thm B], a change of shape is a change of $\hbar F_1$ and of ψ, hence a combination of (a)–(d).*

Proof. (a)–(c): Prop c8-laws, Prop c9-scheme and Prop c11-freedom, with $[\{S^{(1)}(x), -\}, \psi'](O) = \{S^{(1)}(x), \psi'(O)\} - \psi'(\{S^{(1)}(x), O\})$; for $\phi^{(0)} = tS^{(1)}(\Delta)$, $\{S^{(1)}(x), S^{(1)}(\Delta)\}$ is $S^{(1)}$ of $\pm[x, \Delta] = \mp\lambda_x x$ by Lemma c2-components with $S^{(2)}(x, \Delta) = 0$ (Def c2-S2). The Konishi entry: Prop c11-freedom (iii) with $\psi'(B_z) = y\bar{F}_z$, the two functionals having Weyl weight 3, degree 0 and the same quantum numbers. (d) Definition 6.1, Proposition 6.2, Def c11-free, Prop c11-S (ii), Proposition 5.6. (e) Prop c10-shape: only the finite parts change; the finite counterterm and the finite parts of the operators absorb the change, the counterterms being unique up to finite local terms. □

Theorem 6.4 (Invariant content)**.**

- *(i) Under (a), (b) and the coupling direction the representation of $\mathfrak{psl}(4|4)$ induced on the cohomology of $\Lambda^{(0)}$ by $-\Lambda^{(1)}$ changes by an isomorphism; under G_B it is composed with $e^{-a\hbar B}$ (Thm c8-invariants). In particular the one-loop components along Σ, Σ^* and K_b are determined on cohomology up to conjugation, up to the rescaling $1 \mp a\hbar$ of the odd generators by the outer direction, and, through P_1, up to the relations of (d).*

- *(ii) For a rank-one $\xi = u \otimes z$ the class $[Q_1^\xi]$ on the cohomology of Q_0^ξ, its kernel, image and rank are invariant under (a), (b), the coupling direction and G_B (Thm c8-twist); the individual two-body kernels Υ_ξ and the one-body kernels are not.*

- *(iii) The relations of Cor c8-homotopy among Q_1, S_1, D_1, K_1, P_1, M_1, R_1 on cohomology, and the relations $S_1 = -Q_1^{\mathrm{adj}} + (\text{Gram terms})$, $K_1 = -P_1^{\mathrm{adj}} + (\text{Gram terms})$ of Cor c8-gram in a reflection-compatible scheme, are covariant under all the changes of Proposition 6.3 with reflection-compatible parameters (real a in (b), $\bar{a} = -a$ for G_B, Cor c8-gram).*

- *(iv) The spectrum of D_1 on cohomology in a fixed classical weight is invariant (Thm c8-invariants (iii)); with the relations of (iii) it fixes the parts of S_1 and K_1 that multiply the position on cohomology (Prop c11-S (iii), Proposition 5.6 (iii)).*

Proof. Thm c8-invariants, Thm c8-twist, Cor c8-homotopy, Cor c8-gram and Proposition 6.3; each change acts on Λ by a conjugation, by a multiple of $\Lambda^{(0)}$ or by the automorphism $e^{-a\hbar B}$, and on the inner product by the corresponding change of Gram (Cor c8-gram). (iv): Prop c11-S (iii) and Proposition 5.6 (iii). □

7 The twist check and the inner-product test

Throughout this section $\xi = u \otimes z \in S_+ \otimes W$ with $z = e_4$, two-component ${\xi_\alpha}^i = u_\alpha \delta_4^i$, $\bar{\xi} = 0$: the generator Q_-^4 of [BK, before (3.51)] and [CY13, (2.1)] (q1_targets, §2.1). Fix $u' \in S_+$ with $u'^\alpha u_\alpha = 1$. $\Gamma(\xi, \xi) = 0$ in all ten components (Lemma c8-ev).

Definition 7.1 (Arity-two kernels along (ξ, ξ))**.** A defect cycle of the pair (X_a, X_b) along (ξ, ξ) is a chain as in Definition 1.2 with exactly one special edge and either two vertices given by terms of $S^{(1)}(\xi)$ or one vertex given by $S^{(2)}(\xi, \xi) = -\frac{1}{2}\int \big((\xi, \lambda^*), (\xi, \lambda^*)\big)$ (eq c8-twistaction). $\Upsilon_{\xi\xi}(X_a, X_b)$ is the

coefficient of ε^0 of their sum, with the sign of the arity-two component of the flow (Thm c8-L (iv)), plus the finite special vertices of the arity-two counterterm, as in (1.2).

Proposition 7.2 (The twisted one-loop differential on pairs)**.** *Assume the one-loop equivariant master equation through arity two along ξ (arity three and higher along Σ being empty, Lemma o-actiontrunc).*

- *(i) For every local functional O, $\hbar Q_1^\xi(O) = \hbar L_1(O)^{(0)} + Q_1(O) - \frac{1}{2}\hbar L_1(O)^{(2)}(\xi,\xi) - \cdots$ (Def c8-twisted) is a sum over the pairs of letters of the two-body kernel*

$$\Upsilon_\xi(X_a, X_b) - \tfrac{1}{2}\Upsilon_{\xi\xi}(X_a, X_b) + (\textit{arity-zero two-body kernel}) + \cdots$$

  plus one-body terms, $-\{r_\xi(\hbar F_1)|_{q=1}, -\}$ restricted to its parts of arity at most two and the arity-zero one-body terms: the untwisted form of the pair rule [BK, (3.7)–(3.8)].

- *(ii) An arity-k two-body kernel along $(\xi,\dots,\xi)$ of a pair of degree d and weight W has degree $d+1-k$ and Weyl weight $W+\frac{k}{2}$. For pairs of field letters ($d=0$) with outputs of fields only, $k\le 1$; for a pair with one ghost letter and outputs of fields of degree 0, $k=2$. On the pairs of Table 4: $\beta\gamma$, $\beta\beta$, $b\gamma$, βb, bb are arity-one kernels Υ_ξ; bc with b represented by the self-dual field strength is $-\frac{1}{2}\Upsilon_{\xi\xi}$; bc with b represented by λ_+^* is Υ_ξ.*

- *(iii) $S^{(2)}(\xi,\xi)$ has only legs λ_+^* and contributes only to outputs containing λ^*; the field-only part of $\Upsilon_{\xi\xi}$ comes from cycles with two vertices $S^{(1)}(\xi)$ and one special edge.*

- *(iv) $(Q_0^\xi)^2 = 0$ and $Q_0^\xi Q_1^\xi + Q_1^\xi Q_0^\xi = 0$; the class $[Q_1^\xi]$ on the cohomology of Q_0^ξ is invariant under the changes of Theorem 6.4 (ii) (Thm c8-twist).*

Proof. (i) Def c8-twisted and Theorem 3.3 in arities zero and one; in arity two the same classification applies to the graphs with two arguments ξ: a graph of order $\hbar$ meeting O in two letters is a cycle with one special edge and the arguments distributed over the vertices $S^{(1)}(\xi)$, $S^{(1)}(\xi)$ or $S^{(2)}(\xi,\xi)$ on the cycle, or on trees, where they belong to the arity-two flow of $L(O)^{(0)}$ and $L(O)^{(1)}(\xi)$ on the left side of eq c8-rep. (ii) Lemma arity of q1_targets (Prop c7-select applied once per argument): each argument lowers the degree by one and raises the weight by $\frac{1}{2}$; outputs of fields have degree ≥ 0, with c of degree 1. The degrees and weights of the entries are those of Table c11-twist. (iii) Lemma S2 of q1_targets: P has no component on λ^*, and a graph has one field–antifield pairing. (iv) Thm c8-twist. □

Proposition 7.3 (The exact vanishing on the chiral bilinears)**.** *For $w_1, w_2 \in W_{\mathbb{R}}^\perp \otimes \mathbb{C}$ with $\rho(w_1)\xi = \rho(w_2)\xi = 0$, two-component $w_{a\,ij} z^j = 0$, the component along ξ of $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A) = \frac{1}{16}\int \omega\,(w_1^{ij}A_{ij}, w_2^{kl}A_{kl})$ vanishes, in every scheme equivariant for translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ and $\mathfrak{g}$ (Prop c11-plane): in kernel form, the sum over the two orderings of $\Upsilon_\xi(A|_\perp, A|_\perp)$ on (w_1, w_2), contracted with the colour pairing, minus $\{\hbar F_1^{(1)}(\xi), O\}$, is zero. This is the untwisted form of $Q_1(\gamma^I\gamma^J) = 0$ [BK, (3.11), (3.77)].*

Proof. Prop c11-plane and Theorem 3.3 (iii). □

Theorem 7.4 (What the comparison with the holomorphic twist tests)**.** *Let Φ be the quasi-isomorphism of local-operator complexes from the cohomology of Q_0^ξ to the cohomology of the free holomorphic Q_0 on the letters $c, \gamma, \beta, b, \partial_{\dot\alpha}$ of [BK], transported through the equivalence of [ESW, Thm 10.9] (Rem c8-twist), and let D be the difference of the two one-loop quantisations of the twisted theory, the restriction $r_\xi(\hbar F_1)$ of the untwisted equivariant one and the counterterm-free holomorphic one of [BK, (2.42), footnote 22], [W18, Thm 3.4], read through Φ. If $D \in \mathbb{C}\,S^\xi + \operatorname{Im} Q_0^\xi$, then $\Phi \circ [Q_1^\xi] \circ \Phi^{-1}$ equals the class of the one-loop differential of [BK, (3.7), (3.64)–(3.69)] up to a multiple of $[Q_0] = 0$; in particular:*

- *(a) the one-body terms of Q_1^ξ, restricted to the twisted letters, are $a\,Q_0^\xi + [Q_0^\xi, N]$ for a number a and a derivation N of the letters (a field rescaling), which act as 0 on cohomology;*

- *(b) on the pairs from $\{c, \gamma\}$ and on (β, c) the two-body kernels lie, on the twisted letters, in the image of Q_0^ξ plus terms cancelled by the one-body terms on cohomology ([BK, (3.11)]); on (γ, γ) they vanish exactly (Proposition 7.3);*

- *(c) on the pairs $\beta\gamma$, $\beta\beta$, $b\gamma$, βb, bb, bc the two-body kernels, restricted to the twisted letters and reduced modulo $\operatorname{Im} Q_0^\xi$ only (total derivatives of local operators are nonzero in the cohomology of Q_0^ξ and are not quotiented; on the field-only outputs the reduction removes the antiholomorphic derivatives and the holomorphic Dirac component $\partial^{\dot\alpha}\bar\lambda_{\dot\alpha 4}$ of a ∂c output, the latter up to terms with antifields), have the colour structure $(1\otimes \mathrm{ad}_{Y_2}\mathrm{ad}_{Y_1})(-,-)^{-1}$, i.e. $\frac{1}{2} f_{ACD} f_{BCE}$ up to normalisation, and the derivative structure $\partial_{\dot\alpha}(\cdot)\,\partial^{\dot\alpha}(\cdot)$ on the two outputs with $\partial_{\dot\alpha} = u'^{\alpha}\sigma^m_{\alpha\dot\alpha}\partial_m$, the derivatives $u^\alpha\sigma^m_{\alpha\dot\alpha}\partial_m$ being Q_0^ξ-exact on the letters (q1_targets §2.3);*

- *(d) the six coefficients so obtained satisfy, modulo the finite-part shifts of Proposition 6.3 (b), the relations [BK, (3.59)–(3.63)] that follow from the order-$\hbar$ part of $\{Q_1, R_0\} + \{Q_0, R_1\} = 0$ for the tree-level $\mathfrak{psl}(3|3)$ commutant of ξ (Target psl33 of q1_targets), the class-level form of the equality of the factors $\frac{1}{2}$ in [BK, (3.64)–(3.69)];*

- *(e) every class-level statement of [BK], [KvLM], [CL25], [KKLP], [KLLS] about $[Q_1]$ on multi-letter classes is the image under Φ of $[Q_1^\xi]$ computed by the pair rule from the kernels of (c), with the normalisation of $\hbar$ relative to the unit fixed by Φ.*

If $D \notin \mathbb{C}\,S^\xi + \operatorname{Im} Q_0^\xi$ the class maps differ by $[\{D, -\}]$, and (a)–(e) become the computation of that class.

Proof. Thm o-twist (iv) and Rem c8-twist: two quantisations differing by a Q_0^ξ-closed D of degree 0 have class maps differing by $[\{D, -\}]$, which vanishes for $D \in \mathbb{C}\,S^\xi + \operatorname{Im} Q_0^\xi$ ($\{S^\xi, -\} = Q_0^\xi$ and $\{Q_0^\xi \chi, -\} = [Q_0^\xi, \{\chi, -\}]$ up to sign). (a): the one-body terms are brackets with local functionals (Theorem 3.3 (iii), Proposition 7.2 (i)), and the condition on D makes them aQ_0^ξ plus commutators with Q_0^ξ on the twisted letters. (b), (c): the pair rule of [BK, (3.7)–(3.11)], the colour structure of Remark 3.9 with two outputs, and the exactness of the antiholomorphic derivatives, $\{Q, \bar Q^1_{\dot\alpha}\} = P_{-\dot\alpha}$ [BK, (2.5)]. (d): the order-$\hbar$ part of Cor c8-homotopy (ii) at (ξ, R) with $[\xi, R] = 0$ restricted by r_ξ (Prop o-S (iii)). (e): $[Q_1^\xi]$ is determined by its values on representatives, which are polynomials in the letters, through the pair rule. □

Remark 7.5 (The renormalisation freedom of the twist)**.** In terms of Proposition 6.3, the freedoms F0–F5 of q1_targets §4 are: F0 the arity-zero finite counterterms, whose s-exact parts (field countings) act as $Q_1^\xi \mapsto Q_1^\xi + [Q_0^\xi, N_y]$ and whose closed coupling part acts as $-aQ_0^\xi$; F1 the free directions θ_V^ξ, $\theta_\perp^\xi$ of $\hbar F_1^{(1)}(\xi)$, of which the cubic equation leaves the common value θ^ξ, the direction $\theta^\xi S^{(1)}(\xi) = \theta^\xi G_B^{(1)}(\xi)$, which acts as $[Q_0^\xi, \mp\theta^\xi N_{R_\xi}]$ (Thm c8-twist); F2 the arity-two finite counterterm along (ξ, ξ), correlated with F1 by the (ξ, ξ) component of the master equation and entering only outputs with λ^* (Proposition 7.2 (iii)); F3 the finite parts of the operators, $[Q_0^\xi, \psi']$, which move the individual entries, for example $c_K \mapsto c_K - \kappa_B y$; F4 the homotopy data of Φ; F5 the outer direction, $[Q_0^\xi, \mp aN_{R_\xi}]$. Each acts as 0 on the cohomology of Q_0^ξ, so $[Q_1^\xi]$, its kernel, image and rank are scheme independent and the six entries of (c) are scheme dependent; the Konishi entry c_K is not a class quantity since $Q_0^\xi O_\zeta = \kappa_B B_z \neq 0$ (Prop o-konishi).

Remark 7.6 (The relation $S_1 = -Q_1^{\mathrm{adj}} + \text{Gram terms}$)**.** By Cor c8-gram, in a reflection-compatible scheme and on a family closed under the generators considered,

$$S_1 \text{ along } S^\alpha{}_i = -\big(Q_1 \text{ along } Q^i_\alpha\big)^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(S^\alpha{}_i), -\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big],$$
$$S_1 \text{ along } \bar S^{\dot\alpha i} = \big(Q_1 \text{ along } \bar Q_{\dot\alpha i}\big)^{\mathrm{adj}} + \hbar\big[\{S^{(1)}(\bar S^{\dot\alpha i}), -\}, \mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1\big].$$

With the kernels of this note, Q_1 and S_1 are computed from the defect cycles along ξ and η, the position part of S_1 being Q_1 along $\rho(b)\eta$ (Proposition 2.6 (i)) and the intrinsic part Υ^0_η new data; Gram_1 comes from the two-point graphs of Def c8-2pt. On the family formed by a traceless scalar bilinear O at the origin and its descendants $\{S^{(1)}(\xi), O\}$ of Weyl weight $\frac{5}{2}$, where $Q_1(O) = c_Q\{S^{(1)}(\xi), O\}$ (Cor c7-Q) and S_1 on the descendants is c_S times the classical transformation back to O (Cor c7-S for the structure), the relation reads $c_S = \bar c_Q + (g_{5/2} - g_2)$ for the unit generators, with g_w the eigenvalue of $\mathrm{Gram}_{\mathrm{tree}}^{-1}\mathrm{Gram}_1$ in weight w on the family; it is the inner-product test of these kernels, to be run once Gram_1 is computed. The outer direction changes c_Q by $-a$ and c_S by $+a$ and preserves the relation exactly when $\bar a = -a$ (Cor c8-gram).

8 One-loop relations among the components

The relations of this section follow from $L \circ L = 0$ (Thm c8-L (iii), (iv)) and are the gates of the engines (Section 9). Write $\Lambda^{(1)}(x) = \Lambda_0^{(1)}(x) + \hbar\Lambda_1^{(1)}(x)$, with $\Lambda_0^{(1)}(x) = \pm\{S^{(1)}(x), -\}$ (Thm c8-L (iv)) and $\hbar\Lambda_1^{(1)}(x)O$ the component of O along x, and $\Lambda^{(0)} = s + \hbar L_1^{(0)}$.

Lemma 8.1 (On-shell vanishing of exact two-field terms)**.** *Let χ be a colour-singlet local functional of degree -1. The part of $s\chi$ with two field letters and no antifield vanishes on free on-shell plane waves: $|p|^2 = 0$ on the scalars and on the components of $A|_V$, $\rho(p)\lambda = 0$ on the gauginos, $(p, a) = 0$ on the polarisations of $A|_V$.*

Proof. The two-field part of $s\chi$ is $Q\chi_2 + \{I_3, \chi_1\}$ with χ_k the part of χ with k letters; χ_1 is a single adjoint letter contracted with a constant tensor and is not a colour singlet unless it vanishes. χ_2 of degree -1 has one antifield and one field, or c^* and c; Q substitutes $A^* \mapsto -\frac{1}{2}d^*dA$, $\lambda^* \mapsto \not{d}\lambda$, $c^* \mapsto d^*(A^*|_V)$ and $A \mapsto -dc$ (Prop c1-Q), so the field-only part of $Q\chi_2$ is a free field equation times a field, zero on shell. □

Proposition 8.2 (The relations at order $\hbar$)**.** *Let $\mathcal{F}$ be a family of colour-singlet local functionals of degree 0, stable under Λ and under the classical transformations, with $L_1^{(0)} = 0$ on $\mathcal{F}$ (a finite part ψ as in Def c10-hyp (ii)), and let $O \in \mathcal{F}$ with $sO = 0$.*

- *(R1) Arity one. $s\big(\hbar\Lambda_1^{(1)}(x)O\big) = 0$ for every generator x: the one-loop component maps cocycles to cocycles. Along ξ: $s\,Q_1(O) = 0$, an exact identity of local functionals.*
- *(R2) Supercharges. For $\xi, \xi' \in \Sigma$ with $[\xi, \xi'] = \Gamma(\xi, \xi')|_V =: b$,*

$$\big[\Lambda_1^{(1)}(\xi), \Lambda_0^{(1)}(\xi')\big]O + \big[\Lambda_0^{(1)}(\xi), \Lambda_1^{(1)}(\xi')\big]O + \Lambda_1^{(1)}(P_b)O = \Lambda^{(0)}(\chi),$$

for a local χ of degree -1 built from the arity-two component of O along (ξ, ξ'); for O without letters λ and c (the scalar bilinears, the families of scalars) the order-$\hbar^0$ part of χ vanishes, $\{S^{(2)}(\xi, \xi'), O\} = 0$, and the right side is s-exact. For $\xi, \xi' \in S_+ \otimes W$, $b = 0$; in the translation-covariant scheme at $\theta = -\frac{1}{3}$, $P_1 \cong 0$ (Cor c11-D1fam).

- *(R3) Supercharges and special supercharges. For $\xi \in \Sigma$, $\eta \in \Sigma^*$ with $[\xi,\eta] = \frac{1}{2}(\xi,\eta)\Delta + M_{\xi\eta} + R_{\xi\eta}$ (Prop c2-special; $M_{\xi\eta} \in \mathfrak{so}(V)$, $R_{\xi\eta} \in \mathfrak{so}(W_{\mathbb{R}}^{\perp})$),*

$$\begin{aligned}&\big[\Lambda_1^{(1)}(\xi),\Lambda_0^{(1)}(\eta)\big]O + \big[\Lambda_0^{(1)}(\xi),\Lambda_1^{(1)}(\eta)\big]O\\ &\quad + \tfrac{1}{2}(\xi,\eta)\,\hbar\Lambda_1^{(1)}(\Delta)O + \hbar\Lambda_1^{(1)}(M_{\xi\eta})O + \hbar\Lambda_1^{(1)}(R_{\xi\eta})O = \Lambda^{(0)}(\chi'),\end{aligned}$$

  with χ' s-exact on families without letters λ, c; on the scalar bilinears $\hbar\Lambda_1^{(1)}(\Delta)O = D_1(O)$ is 0 (traceless) and $-48u\,O$ (trace) at $\theta = -\frac{1}{3}$ (Cor c9-ident), $R_1 = 0$ (Rem c7-internal), and $M_1(O)$ is the position part $(\omega x)^\mu P_1$ of the rotation component, $\cong 0$ in the covariant scheme.

- *(R4) Twisted closure. For a rank-one ξ, $Q_0^\xi Q_1^\xi + Q_1^\xi Q_0^\xi = 0$ on all local functionals, an exact identity (Thm c8-twist).*

- *(R5) Special conformal and translations. $\big[\Lambda_1^{(1)}(K_b),\Lambda_0^{(1)}(P_{b'})\big] + \big[\Lambda_0^{(1)}(K_b),\Lambda_1^{(1)}(P_{b'})\big] + (b,b')\hbar\Lambda_1^{(1)}(\Delta) + \hbar\Lambda_1^{(1)}(M_{bb'}) \cong 0$ on cohomology, $[K_b,P_{b'}] = (b,b')\Delta + M_{bb'}$ (Rem c7-KDelta); with Proposition 5.6 (i) this is the statement that the parts of K_1 multiplying the position are those of* (5.2).

- *(R6) Two special supercharges. For $\eta_1,\eta_2 \in \Sigma^*$ with $[\eta_1,\eta_2] = -K_b$, $b = \Gamma(\eta_1,\eta_2)|_V$: K_1 along K_b is, on cohomology, $\big[\Lambda_1^{(1)}(\eta_1),\Lambda_0^{(1)}(\eta_2)\big] + \big[\Lambda_0^{(1)}(\eta_1),\Lambda_1^{(1)}(\eta_2)\big]$ (anticommutators), on families without letters λ, c (where $\{S^{(2)}(\eta_1,\eta_2),O\} = 0$) and at finite T.*

In (R2), (R3), (R5), (R6) the right sides vanish on cohomology; on colour-singlet families without λ, c, their two-field, antifield-free parts vanish on free on-shell plane waves (Lemma 8.1).

Proof. Cor c8-homotopy (i) at order $\hbar$: $[s,\Lambda_1^{(1)}(x)] + [L_1^{(0)},\Lambda_0^{(1)}(x)] = 0$; on O with $sO = 0$, $L_1^{(0)}O = 0$ and $L_1^{(0)}\{S^{(1)}(x),O\} = 0$ (stability of $\mathcal{F}$) this is (R1). Cor c8-homotopy (ii) at order $\hbar$ for the pairs (ξ,ξ'), (ξ,η), $(K_b,P_{b'})$, (η_1,η_2), with the brackets of Prop c2-special and Rem c7-KDelta: the right side is $[\Lambda^{(0)},H]$ with H built from $\Lambda^{(2)}$, and on O with $\Lambda^{(0)}O = 0$ it is $\Lambda^{(0)}(HO)$; the order-$\hbar^0$ part of H is $\pm\{S^{(2)}(x_1,x_2),-\}$, and $S^{(2)}$ has the legs (λ^*,λ^*) and (A,c^*) (Def c2-S2), whose bracket with a functional of the fields A alone vanishes. (R4): Thm c8-twist. The vanishing on shell: Lemma 8.1, with $L_1^{(0)} = 0$ on $\mathcal{F}$. □

Lemma 8.3 (The trace bilinear in a cocycle scheme)**.** *Let $O = \int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, $F_1 = \{S^{(1)}(\xi),O\}$, and F_2, F_3, F_4 the trace functionals $(w_1,w_2)\int\omega\,(\rho(A|_\perp)\xi,\lambda)$, $(w_1,w_2)\int\omega\,(\rho(A|_V)\xi,\lambda)$, $(w_1,w_2)\int\omega\,(c,(\xi,\lambda^*))$ of Cor c7-Q.*

- *(i) If $L_1^{(0)} = 0$ on O and on F_1, the two-body coefficients a_3, a_4 of F_3, F_4 in $Q_1(O)$ vanish: $Q_1(O) = a_1F_1 + a_2F_2$ plus one-body terms, which are multiples of F_1.*

- *(ii) If $L_1^{(0)}(O) = 2\kappa\,(w_1,w_2)\int\omega\,(dc,A|_V)$ and the two-field part of $L_1^{(0)}(F_1)$ is $-\kappa'\,sF_3$ at $\psi = 0$, the finite part $\psi(O) = \kappa\,(w_1,w_2)\int\omega\,(A|_V,A|_V)$, $\psi(F_1) = \kappa'F_3$ removes them in the two-field sector, and changes $Q_1(O)$ by $(2\kappa-\kappa')F_3$; then (R1) at $\psi = 0$ is the identity $a_3 + 2\kappa - \kappa' = 0$ on the two-field sector (c,λ).*

Proof. (i) By (R1), $s\,Q_1(O) = 0$. $sF_1 = 0$, since $[s,\{S^{(1)}(\xi),-\}] = \{\{S,S^{(1)}(\xi)\},-\} = 0$ at tree level and $sO = 0$; $sF_2 = 0$, since Q annihilates $A|_\perp$ and λ on functionals and the cubic parts of s on $A|_\perp$ and λ are the gauge transformation with parameter c, which leaves the invariant pairing fixed. In the two-field sector (c,λ), $sF_3 = -(w_1,w_2)\int\omega\,(\rho(dc)\xi,\lambda)$ and $sF_4 = (w_1,w_2)\int\omega\,(c,(\xi,\not{\partial}\lambda))$, whose densities carry the derivative on different letters and are linearly independent for ω arbitrary; hence $a_3 = a_4 = 0$.

The one-body terms are $-\{\hbar F_1^{(1)}(\xi), O\}$, and $S_2^V(\xi)$ does not act on $A|_\perp$. (ii) $Q\int\omega\,(A|_V, A|_V) = -2\int\omega\,(dc, A|_V)$ (Rem c5-cocycle), and the cubic part of $s\int\omega\,(A|_V, A|_V)$ vanishes by invariance of $(-,-)$; L_1 changes by $[s,\psi]$ (Lemma c10-psi), the component along ξ by $[\{S^{(1)}(\xi),-\},\psi]$ (Prop c8-laws (b)), and $\{S^{(1)}(\xi), \int\omega\,(A|_V, A|_V)\} = 2\int\omega\,(\Gamma(\xi,\lambda)|_V, A|_V)$ with $(\Gamma(\xi,\lambda)|_V, a) = (\rho(a)\xi, \lambda)$ [ESW, (28)]. In the shifted scheme (i) gives $a_3 + 2\kappa - \kappa' = 0$; the two-field part of (R1) at $\psi = 0$ is the same equation, since $\{S^{(1)}(\xi), L_1^{(0)}(O)\} = 2\kappa\, sF_3$ and $L_1^{(0)}(F_1) = -\kappa' sF_3$ there. □

Remark 8.4 (The scalar bilinears as the first test family)**.** On the scalar bilinears every relation of Proposition 8.2 is a finite computation with the two-body kernels of the pairs $(A|_\perp, A|_\perp)$ along ξ, η, K_b and of $(A|_\perp, \lambda_\pm)$ along ξ, η, the one-body kernels of $A|_\perp$ and λ, and the known D_1, P_1 (Thm c9-D1, Thm c9-P1, Prop c9-scheme): (R1) is $s\,Q_1(O) = 0$; (R3) on the trace bilinear is the Konishi relation $\{Q_1, S_0\} + \{Q_0, S_1\} \cong -\frac{1}{2}(\xi,\eta)(-48u)$ on the two-field part, and 0 on the traceless part; (R2) with $\xi \in S_+ \otimes W$, $\xi' \in S_- \otimes W^*$ is $\cong -P_1(b) = (\frac{1}{3} + \theta_P)u\,\partial_b$, which vanishes in the covariant scheme. By Cor c7-Q, Cor c7-S and Cor c7-K the components have the forms $Q_1(O) = c_Q\{S^{(1)}(\xi), O\}$ on the traceless part, three coefficients on the trace part, S_1 of the same form at $\rho(x)\eta$, and $K_1 \cong -(b,x)D_1$; the relations fix c_S and the trace coefficients in terms of c_Q, up to the outer direction. The hypothesis $L_1^{(0)} = 0$ on the family does not hold at $\psi = 0$: on the trace bilinear $L_1^{(0)}(O)$ is a nonzero multiple of $(w_1, w_2)\int\omega\,(dc, A|_V)$ (Rem c5-cocycle), and the relations are tested after the finite part of Lemma 8.3 (ii), with which only F_1 and F_2 remain on the trace part; this uses in addition the arity-zero kernels of $(A|_\perp, A|_\perp)$ and $(A|_\perp, \lambda_+)$.

9 Engine specification

The machine-readable specification is `step2/composite_qsk/qsk/theory/engine_spec.json`; this section states its content. Both routes compute the same objects, the two-body kernels Υ_x of Definition 1.2 with their pole and finite parts and the one-body inputs, by independent integration methods and independently written numerator algebra.

Route A (exact simplex integrals, extending `step2/twoletter/engine_a` and `step2/composite_qsk/composite/engine_a`): chain numerators from the vertex functions of I of the two-letter engine A and the terms of $S^{(1)}(x)$ of Table 1 realised as in `superconformal/model.py`; the special edge with Schwinger time in $[0,\varepsilon]$ and numerator $-|l+q_\sigma|^2$; the Gaussian loop integral and Theorem 5.2 with exact Dirichlet–log integrals over the ordered subsimplices of the region where α_σ is smallest; letter derivatives by Proposition 5.3; the injected momentum r and its derivatives along η and K_b; colour stripping and the planar adjacent reading of Remark 3.9. Values in $\mathbb{Q}[i]$ combinations of 1, $\log 2$, $\log 3$ and, for $n \geq 3$ vertices, the basis of the Part-C engines, confirmed by a second exact evaluation.

Route B (momentum space, extending `step2/twoletter/engine_b` and `step2/composite_qsk/composite/engine_b`): explicit momentum-space numerator algebra from an independently written realisation of $S^{(1)}(x)$ (the Clifford data of the two-letter engine B with the field transformations of Chapter 2 written out anew), the special edge as the explicit factor $e^{-\varepsilon|p|^2} - 1$, Feynman parameters for the other edges, numerical evaluation at several ε with ≥ 30 digits and Richardson extrapolation to ε^0, the poles fitted and compared; along K_b at the finite endpoints $T|p|^2 \in \{1, 4, 16\}$; and, on the scalar bilinears and the Konishi descendant, the second evaluation of Rem c9-twoeval: Q applied to the arity-one part of $O[\infty]$ along ξ (Lemma c7-image), the graphs with $S^{(1)}(\xi)$ in the flow on a leg, on the cycle and on a one-loop subgraph of a leg, whose sum must be local and equal to (3.1).

Tasks. E0 conventions (the commutators of Proposition 2.2 on plane waves, the arrows of Lemma 2.1, the sign rule of Definition 1.2 (ii) fixed against G-cubic and G-pole, the canonical form with $\gamma^{\mu\nu}\partial_\mu\partial_\nu\delta$, the twisted coefficient $-\frac{1}{2}$); TQ0 the quadratic free directions of Proposition 6.2 (i) and their analogues along η and K_b; TQ1 the action side along ξ: the quadratic part of $\mathrm{Ob}^{(1)}(\xi)$ in the sectors

$(\lambda, A|_V)$, $(\lambda, A|_\perp)$, (c, λ^*), the bracket $\{S^{(1)}(\xi), \hbar F_1^{(0)}\}$ with the finite counterterm of Thm c4-finite and its β-family, and the solution $\hbar F_1^{(1)}(\xi)$ of (6.1) with its family θ_V^ξ, $\theta_\perp^\xi$; TQ2 the kernels Υ_ξ on the ten pairs of field and ghost letters, $n \le 3$ derivatives (route B $n \le 2$), all leg sets of Table 2 and of the census file, with the finite special vertices; TQ3 the pairs of Table 4 at $\xi = u \otimes e_4$ (letter vectors with non-integral components contracted exactly) and the decomposition of Theorem 7.4 (c), including the arity-two kernel $\Upsilon_{\xi\xi}$ of the entry bc; TQ4 the scalar bilinears along ξ: c_Q and the three coefficients of the trace part (Cor c7-Q), with their scheme families; TQ5 along η: the position identity and Υ_η^0 on $(A|_\perp, A|_\perp)$, $(A|_\perp, \lambda_\pm)$, and S_1 on the bilinears and their weight-$\frac{5}{2}$ descendants; TQ6 along K_b: $\Upsilon_{K_b}^0$ for $n \le 2$ and K_1 on the bilinears and the letters by (5.2); TQ7 the regression along Δ and P_b by the route of the special edges; TQ8 (after the two-point functions) the inner-product tests of Remark 7.6 and Remark 5.7.

Gates. G-E0 the commutators; G-TC0 the free directions; G-cubic Example 5.5; G-noLog no $\log \varepsilon$ in any defect cycle; G-pole (3.3) and G-log (3.2) pair by pair at $n \le 1$, with the vacuum-pole terms of Theorem 3.3 (iv); G-pos the position identities of Proposition 2.6; G-reg Cor 3.6 and the first row of the table of Thm c9-D1 ($4\,\mathrm{lc} - \frac{4}{3}O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ on the bilinears); G-plane Proposition 7.3; G-R1 $sQ_1(O) = 0$ on the bilinears, at $\psi = 0$ in the form of Cor c8-homotopy (i) with the Grassmann order of the outputs (Lemma 8.3 (ii)); G-R2, G-R3 (the Konishi relation through D_1), G-R5, G-R6 on the bilinears on shell (Lemma 8.1); G-R4 the twisted closure on the pairs of Table 4; G-KT the independence of T of Υ_{K_b} (route B at three endpoints on a defect cycle along K_b, including one with the $A|_V$–c line on the cycle; route A by construction) and the vanishing of the $\log T$ coefficient of the defect graphs as $\varepsilon \to 0$; G-KD $K_1 \cong -(b, x)D_1$ on the bilinears; G-twist Theorem 7.4 (a)–(d), reductions modulo $\operatorname{Im} Q_0^\xi$ only; G-Ob the quadratic $\mathrm{Ob}^{(1)}(\xi)$ lies in the image of (6.1); G-AB agreement of the routes entry by entry; G-colour the planar rule in the realisations $\mathfrak{su}(2)$, $\mathfrak{su}(3)$, $\mathfrak{gl}(3)$.

10 Two-component forms

Remark 10.1 (Weinberg–Srednicki two-component forms)**.** With the conventions of Def c8-twocomp (Euclidean $\sigma^m = (\sigma^1, \sigma^2, \sigma^3, i)$, $\bar\sigma^m = (-\sigma^1, -\sigma^2, -\sigma^3, i)$, $\sigma^m\bar\sigma^n + \sigma^n\bar\sigma^m = -2\delta^{mn}$, $\varepsilon^{12} = \varepsilon^{\dot1\dot2} = +1$, $\chi\psi = \chi^\alpha\psi_\alpha$, $\bar\chi\bar\psi = \bar\chi_{\dot\alpha}\bar\psi^{\dot\alpha}$):

- Generators. $\xi = (\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$ along Q_α^i, $\bar Q_{\dot\alpha i}$; $\eta = (\eta^\alpha{}_i, \bar\eta^{\dot\alpha i})$ along $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$; the Killing spinor $\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$ (eq c8-killing); K_b: $v^m = x^2 b^m - 2(b\cdot x)x^m$, $\operatorname{div} v = -8\,b\cdot x$, $\partial_m v_n - \partial_n v_m = 4(x_m b_n - x_n b_m)$. The one-loop components are $Q_1(O) = \xi_\alpha{}^i Q_1(O)^\alpha{}_i + \bar\xi_{\dot\alpha i}\bar Q_1(O)^{\dot\alpha i}$, $S_1(O) = \eta^\alpha{}_i S_1(O)_\alpha{}^i + \bar\eta^{\dot\alpha i}\bar S_1(O)_{\dot\alpha i}$ (Def c8-components).

- Vertices. $S^{(1)}(\xi)$ for $\xi_\alpha{}^i = u_\alpha z^i$ is the sum of the terms of Table 1 (Rem c8-twist); for general $(\xi, \bar\xi)$ the terms with $\bar\xi$ are their conjugates under $\lambda_\alpha{}^i \leftrightarrow \bar\lambda_{\dot\alpha i}$, $\lambda^{*\alpha}{}_i \leftrightarrow \bar\lambda^{*\dot\alpha i}$, $\sigma \leftrightarrow \bar\sigma$. $S^{(1)}(\eta) = S^{(1)}(\xi)|_{\xi\to\epsilon(x)} + \int\big\{(A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \lambda^{*\alpha}{}_k) - (A_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}, \bar\lambda^{*\dot\beta k})\big\}$. $S^{(1)}(K_b) = \int\big\{(v.A_m, A^*_m) + \frac14(v.A^{ij}, A^*_{ij}) - (v.\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (v.\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (v.c, c^*)\big\}$ with $v.A_m = v^n\partial_n A_m + A_n\partial_m v^n$, $v.A_{ij} = v^n\partial_n A_{ij} - 2(b\cdot x)A_{ij}$, $v.\lambda_\alpha{}^i = v^n\partial_n\lambda_\alpha{}^i - \frac12(x_m b_n - x_n b_m)(\sigma^m\bar\sigma^n)_\alpha{}^\beta\lambda_\beta{}^i - 3(b\cdot x)\lambda_\alpha{}^i$, $v.\bar\lambda_{\dot\alpha i} = v^n\partial_n\bar\lambda_{\dot\alpha i} + \frac12(x_m b_n - x_n b_m)\bar\lambda_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} - 3(b\cdot x)\bar\lambda_{\dot\alpha i}$, $v.c = v^n\partial_n c$ (Def c7-vdot, Lemma c7-weights).

- Commutators. On a plane wave, C_ξ maps λ to c by $\lambda \mapsto -p_m\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big)$, the image of $+i(p, \Gamma(\xi,\lambda)|_V)$ with $\Gamma(\xi,\lambda)|_V^m = i(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i)$ (Def c8-twocomp). For η: $\xi_m = \rho(e_m)\eta$ has the components $(\xi_m)_\alpha{}^i = -i\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $(\xi_m)_{\dot\alpha i} = i\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, and $[\eta., H] = 2\sum_m \xi_m.\,\partial_m$. For K_b, (2.5) with $\Sigma^{\mu\nu}$ acting on A_m as $(\Sigma^{\mu\nu}A)_m = \delta_{m\mu}A_\nu - \delta_{m\nu}A_\mu$ and on $\lambda_\alpha{}^i$ through $-\frac14(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu)$.

- Composite line along ξ. For $\xi_\alpha{}^i$, the line joins a leg $\lambda_\alpha{}^i$ to a leg c through $\langle\lambda_\alpha{}^i\bar\lambda_{\dot\beta j}\rangle$, the term

T_1 and $\langle A^*_m c\rangle$; with $\sigma^k_{\alpha\dot\beta}\bar\sigma^{m\dot\beta\gamma}p_k p_m = -|p|^2\delta^\gamma_\alpha$ it is a multiple of $\xi_\alpha{}^i\, e^{-2\varepsilon|p|^2}|p|^{-2}$ times $(-,-)^{-1}$ at $T=\infty$: a scalar line of momentum degree -2 carrying the constant spinor ξ. Both engines fix the multiple (task E0): $2(P\star\,\xi.\,P\star\otimes 1)K_0$ is $-2\,\xi\, e^{-2\varepsilon|p|^2}|p|^{-2}$ on $c\otimes\lambda_+$ and $+2\,\xi\, e^{-2\varepsilon|p|^2}|p|^{-2}$ on $\lambda_+\otimes c$, real.

- Kernels. Υ_ξ of a pair of two-component letters is the ε^0 coefficient of the contraction $\langle X_a(x)X_b(y)\rangle$ at $y=x$ in the background of the fields, with one special edge, built from the propagator components of Prop c8-twocomp, the vertices of Prop c8-twocomp and one vertex of $S^{(1)}(\xi)$; the special edge carries $e^{-\varepsilon|p|^2}-1$ times $(-,-)^{-1}$ and the inverse fibre pairing of (A_m, A^*_m), $\frac14(A^{ij}, A^*_{ij})$, (λ,λ^*), (c,c^*) (Rem c11-twocomp). Along $S^\alpha{}_i$ the position part along b is the kernel along $\bar Q_{\dot\alpha i}$ with $\bar\xi_{\dot\alpha i} = i\,b_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, along $\bar S^{\dot\alpha i}$ the kernel along Q^i_α with $\xi_\alpha{}^i = -i\,b_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$.

- Twist. $\xi_\alpha{}^i = u_\alpha\delta^i_4$; the letters $\gamma^I = A_{I4}$, $\beta_I = u'^\alpha\lambda_\alpha{}^I$ $(I=1,2,3)$, $\partial_{\dot\alpha}c = \bar\lambda_{\dot\alpha 4}$, c, $b = u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta$ or the component $u_\alpha\lambda^{*\alpha}{}_4$ of λ^*_+ (the component of the pair (c,λ^*_+) that reaches the output $\bar\lambda_{\dot\alpha 4}\bar\lambda^{\dot\alpha}{}_4$ by $SU(3)\times U(1)$ and Lorentz invariance), $\partial_{\dot\alpha} = u'^\alpha\sigma^m_{\alpha\dot\alpha}D_m$. The Konishi entry is the coefficient c_K of $\int\omega\,(\bar\lambda_{\dot\alpha 4},\bar\lambda^{\dot\alpha}{}_4)$ in $Q_1\big(\int\omega\,(u'^\alpha\lambda_\alpha{}^i, A_{i4})\big)$; the chiral bilinears of Proposition 7.3 are $\int\omega\,(m^i_1 A_{i4}, m^k_2 A_{k4})$; [BK, (3.66)] is $Q_1(\beta^A_I\gamma^{J\,B}) = \frac12\delta^J_I f_{ACD}f_{BCE}\,\partial_{\dot\alpha}c^D\partial^{\dot\alpha}c^E$.

- Weinberg. On the Dirac column $(\lambda_\alpha{}^i, \bar\lambda^{\dot\alpha}{}_i)$, $\rho(v) = \gamma^\mu_W v_\mu$ gives Weinberg's form of each line (Prop c8-twocomp); with $A_m = -igA^a_m T^a$, $A_I = -igA^a_I T^a$, $\lambda = -\frac{ig}{\sqrt2}\lambda^a T^a$, $\mathrm{tr}(T^aT^b) = \frac12\delta^{ab}$, the unit is $-\frac{g^2N}{32\pi^2}$.

11 Status

Proved here: the classification of the defect graphs along every generator and the two-body form (3.1) for every local functional (Theorem 3.3), with the existence of the local representatives through arity one given the master equation through arity one (Corollary 3.4); the commutators of the generators with the regulator and the composite lines and variation kernels they produce (Proposition 2.2, Corollary 2.3); the Ward identity of the flow along Σ (Proposition 2.4); the position decomposition of the kernels along Σ^* and K_b (Proposition 2.6); the output bounds (Proposition 4.1); the simplex form, the absence of $\log\varepsilon$ and the independence of T along K_b (Theorem 5.2); the pole identity and the arity-one logarithmic covariance (Theorem 3.3 (iv)); the transformation laws and the invariant content (Proposition 6.3, Theorem 6.4); the exact vanishing on the chiral bilinears (Proposition 7.3); the relations of Proposition 8.2. Computed here: the census (Proposition 4.2, Tables 2–4, 0 violations). The numbers of the kernels, of the action side along Σ and of the twist entries are the tasks of the engines; the comparison with [BK] at the level of classes requires the transport map Φ of Theorem 7.4, derived from the collections of [ESW, Thm 10.9], and the inner-product tests require the two-point functions of Def c8-2pt.

Revision of 2026-10-30, after the comparison of the two engine routes: the sign of Lemma 2.1 (now $\lambda\mapsto+\Gamma(\xi,\lambda)$, $A\mapsto+\frac12\rho(dA)\xi$, so that $\{S^{(1)}(\xi),O\} = +F_1$ on the scalar bilinears) and its consequences in Proposition 2.2 (i), Example 5.5, Proposition 6.2 (iii) and Remark 10.1; the restriction of the components of C_ξ and of $\mathcal{V}_\xi$ to V (Proposition 2.2 (i), Corollary 2.3 (i)); the constant arrows Y_η in η. (Proposition 2.2 (ii)); the closed quadratic directions along Σ^* (Proposition 6.2); the vacuum-pole terms of the pole identity read pair by pair (Theorem 3.3 (iv)); the trace bilinear in a cocycle scheme (Lemma 8.3, Remark 8.4); the reduction of the twisted entries modulo $\mathrm{Im}\,Q^\xi_0$ only (Theorem 7.4 (c)); the real composite line along ξ (Remark 10.1). The computed kernels, gates and the twist entries are collected in `step2/composite_qsk/results/qsk_final.json`.